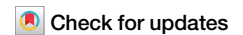


<https://doi.org/10.1038/s42005-025-02447-w>

Observational causality by states and interaction type for scientific discovery



Álvaro Martínez-Sánchez ¹✉ & Adrián Lozano-Durán ^{1,2}

Causality plays a central role in understanding interactions between variables in complex systems. These systems often exhibit state-dependent causal relationships, where both the strength and direction of causality vary with the value of the interacting variables. In this work, we introduce a state-aware causal inference method that quantifies causality in terms of information gain about future states. The effectiveness of the proposed approach stems from two key features: its ability to characterize causal influence as a function of system state, and its capacity to distinguish between redundant and synergistic interactions. The method is validated across a range of benchmark cases in which the direction and strength of causality evolve in a prescribed manner with the state of the system. We further demonstrate the applicability of our approach in two real scenarios: the interaction between motions across scales in a turbulent boundary layer, and the Walker circulation phenomenon in tropical Pacific climate dynamics. Our results show that, without accounting for state-dependent causality as well as redundant and synergistic effects, traditional approaches to causal inference may lead to incomplete or misleading conclusions.

Causality is a fundamental concept in understanding the interactions between variables in complex systems. It provides insights into how one variable influences another and guides the implementation of meaningful changes within these systems¹. In many real-world scenarios, the nature of these interactions is not uniform across all states of the system; they often vary depending on the value of the variables involved. For example, in climate science, the direction and magnitude of causality between atmospheric variables can shift dramatically under different conditions, such as the wind flow direction in the Pacific Region during El Niño or La Niña events². Similarly, in neuroscience, the causality among brain signals often depends on the activation intensity of surrounding regions³. These examples highlight the importance of developing methods that not only detect causal relationships on average, but also capture how these interactions vary with the state of the system—and consequently, over time. By addressing this need, state-aware causal analysis can drive progress in a wide range of scientific and engineering fields, including climate science⁴, neuroscience⁵, economics⁶, epidemiology⁷, social sciences⁸, and fluid dynamics^{9,10}.

To date, most causal inference methods offer an estimate of the average causal strength across all system states, without providing insight into how causality varies at the level of individual states. One such method is convergent cross mapping (CCM)¹¹ and its variants^{12–15}, which infer causality by analyzing connections in the reconstructed attractor of the underlying dynamical system using different combinations of

observed variables. A similar limitation arises in the Peter–Clark Momentary Conditional Independence (PCMCI) framework¹⁶ and its extensions^{17,18,93}, which focus on identifying a minimal conditioning set that includes the causal parents of the target variable. Recent advances in causal representation learning¹⁹ have demonstrated improved performance over PCMCI; nevertheless, their estimates still reflect an average causal strength aggregated across all system states. Traditional approaches, such as Granger causality²⁰, also suffer from this limitation, as they assess causality by testing whether the past values of a source variable improve the prediction of a target variable in the future, but only in an average sense over all system states. Feature-ranking methods, such as the distributed information bottleneck²¹, have gained popularity for isolating the most informative variables in systems to enhance predictive accuracy. However, these approaches do not explicitly address causality, let alone how causal influence may vary with the state of the system.

Among the methods that offer a decomposition of causality by states, one of the most intuitive formulations relies on the concept of interventions^{22,23}. Interventions actively modify one variable to observe its effect on another. A family of approaches within this framework is based on the concept of effective information^{24–26}, which quantifies the strength of causal interactions by measuring the change in the state of the system when a variable is replaced with maximally uncertain noise. These methods provide a state-dependent decomposition of causality by selectively intervening under specific conditions (e.g., intervening on a variable A only when $A = 1$

¹Department of Aeronautics and Astronautics, Massachusetts Institute of Technology, Cambridge, MA, USA. ²Graduate Aerospace Laboratories, California Institute of Technology, Pasadena, CA, USA. ✉e-mail: alvaroms@mit.edu

to measure the causal effect from that state). However, in many real-world settings, directly manipulating a system to test cause-effect relationships is often not feasible. Practical limitations, technical constraints, and the risk of disrupting the underlying dynamics can make controlled interventions extremely difficult to implement. In addition, in some domains, such as neuroscience or climate research, actively perturbing the system may also be inappropriate or ethically unacceptable¹⁶. Moreover, even when interventions are theoretically possible, defining which intervention is appropriate and assessing whether it would preserve the natural behavior of the system remains an open challenge, as external forcing may push the dynamics away from the attractor on which the system typically evolves.

Information theory²⁷ has also become a foundational framework for quantifying causality. The motivation for employing information-theoretic approaches stems from the recognition of information as a fundamental property of physical systems^{28,29}. Early applications of information-theoretic causality involved the use of conditional entropies to define directed information^{30,31}. One of the most recognized developments in this area is the introduction of transfer entropy (TE)³², which quantifies how knowledge of the past states of one variable reduces uncertainty about the future state of another. Subsequent efforts to refine TE led to the development of conditional transfer entropy (CTE)^{33–37}, aimed at addressing multivariate analyses. Other information-theoretic approaches, inspired by dynamical systems theory^{38–41}, quantify causality as the amount of information flowing from one process to another, as dictated by the governing equations of the system. Similar to previous methods, these information-based approaches provide only an average measure of causal strength and do not offer insight into which specific states contribute most to the observed causality.

Several extensions of TE have been developed to account for state-dependent information transfer⁴², although many are formulated primarily to quantify information dynamics rather than to perform causal inference. One of these extensions is local TE^{34,36,43}, which evaluates the information transfer between specific states of the (past) source and (future) target variables by analyzing the expectation term in the original TE formulation. This method preserves several desirable properties, including the ability to recover the standard TE when averaged over all states. Local TE can take negative values, reflecting a combination of contributions from both informative states (positive values) and misinformative states (negative values). Specific TE was later introduced⁴⁴, refining the interpretation of state-dependent information transfer by projecting local TE onto the states of the target variable. This projection yields a non-negative decomposition, enhancing interpretability; however, it considers only the past states of the source variables. Other approaches, such as ensemble TE^{45–49} or time-varying Liang–Kleeman information transfer^{50,51}, explore information transfer across varying time windows to detect shifts in causal structure over time. More recently, a local version of Granger causality⁵²—viewed as the linear counterpart to local TE—has been proposed. This formulation offers a computationally efficient alternative for analyzing state-dependent causal interactions in linear systems with limited data.

While the methods above have advanced our understanding of state-dependent interactions, they are unable to distinguish between synergistic, unique, and redundant causal influences. It has been shown that the inability to disentangle these components can obscure the true causal structure of a system, often leading to spurious or misleading conclusions⁵³. For example, consider the effect of two genes, *A* and *B*, on the expression of eye color, *C*. These interactions can manifest in several ways: *synergistic*, where both genes must be active together to produce a specific eye color—such as green—that neither gene can induce on its own; *unique*, where gene *A* contributes to blue eye color via a pathway independent of gene *B*; and *redundant*, where either gene alone is sufficient to produce brown eyes due to overlapping biological pathways. Without a method capable of isolating these different types of causal contributions, such complexities remain hidden, and critical insights into causal relationships are missed.

The first attempts to disentangle synergistic, unique, and redundant components were introduced in information theory through the partial

information decomposition (PID) proposed by Williams and Beer⁵⁴. However, PID was not originally developed for causal inference, and its use in this context has remained limited. Recently, the synergistic-unique-redundant decomposition of causality (SURD)⁵⁵ was introduced, an information-theoretic method for causal inference that explicitly accounts for the distinct contributions of synergistic, unique, and redundant causal influences, although its definitions of redundancy and synergy differ from those in PID. Other information-theoretic approaches have also been proposed to separate synergistic and redundant effects within TE^{55,56}. Nevertheless, neither SURD nor these related methods were designed to capture how causal influences vary with the state of the system. As a result, these methods cannot distinguish between scenarios in which a specific value of gene *A* = *a'* exerts a unique causal influence on a particular state *c'* of the condition *C*, and those in which its effect is redundant or synergistic with a specific value '*b*' of gene *B*.

In this work, we present a causal inference method that simultaneously accounts for synergistic, unique, and redundant causal influences, while also decomposing these contributions at the level of individual system states. We show that this approach outperforms existing causal inference methods, including the original SURD formulation, by providing a more detailed, state-aware characterization of causal interactions. We validate the method across a range of benchmark problems in which causal pathways depend on the system state in a controlled and known manner. We also demonstrate the applicability of our approach in two real-world scenarios—fluid dynamics and climate science—highlighting its ability to uncover state-specific causal relationships.

Results

State-dependent and interaction-specific causality

Consider the collection of *N* time-evolving source variables given by the vector $\mathbf{Q} = [Q_1(t), Q_2(t), \dots, Q_N(t)]$. For example, Q_i may represent the activity level of a specific brain region over time, a binary variable indicating the presence or absence of an event (e.g., whether it rained on a particular day), or the daily average of a stock market index (e.g., S&P 500). The components of $\mathbf{Q}$ are observables and are treated as random variables. A particular realization of $\mathbf{Q}$ is denoted by $\mathbf{q}$, representing a specific state (i.e., value). Following the examples above, q_i may correspond to a particular level of neural activity, one of the two states (0 or 1) of the binary variable indicating whether it rained or not, or a specific value of the stock market index (e.g., the closing value on a given day). These states are defined by the user according to the specific requirements of the problem. Our objective is to quantify the causal influence of the states of the source, $\mathbf{q}$, on the future state of a target variable, denoted as $q_j^+ = q_j(t + \Delta T)$, where $\Delta T > 0$ represents the future time interval over which causal influence is assessed. The variable Q_j is often included in $\mathbf{Q}$, but this is not required in general.

Our approach is formulated in three steps. First, following the principle of *forward-in-time information propagation*—i.e., information flows only toward the future²⁹—we quantify causality among variables in terms of increments of information. Second, we separate these increments into causal and non-causal components. Third, we further decompose the causal increments into distinct types of interactions: synergistic, unique, and redundant contributions.

For step one, we adopt the definition of causality proposed in ref. 53. In this framework, causality is quantified as the increase in information about each future state q_j^+ gained by observing individual or groups of past states $\mathbf{q}$. The information content in Q_j^+ is measured by Shannon information (or entropy)²⁷, denoted as $H(Q_j^+)$, which represents the average number of bits needed to determine the value of Q_j^+ unambiguously. Shannon information can also be interpreted as a measure of uncertainty: highly uncertain processes (high entropy) yield greater information gain when their outcomes are revealed, whereas completely deterministic processes (zero entropy) provide no new information upon observation.

For the second step, we decompose the information in $H(Q_j^+)$ into a sum of information increments contributed by each past state of the

observable vector $\mathbf{Q}$:

$$H(Q_j^+) = \sum_{q_j^+ \in Q_j^+} \sum_{\mathbf{q} \in \mathbf{Q}} [\Delta C(q_j^+; \mathbf{q}) + \Delta \mathcal{N}(q_j^+; \mathbf{q})] + \Delta I_{\text{leak} \rightarrow j}, \quad (1)$$

where the terms $\Delta C(q_j^+; \mathbf{q}) > 0$ and $\Delta \mathcal{N}(q_j^+; \mathbf{q}) \leq 0$ represent the causal and non-causal contributions, respectively, for a given source and target state. The term $\Delta I_{\text{leak} \rightarrow j}$ is referred to as the causality leak²⁹. The mathematical expression for each of the terms in Eq. (1) is given in Methods and its derivation is detailed in Supplementary Note 1.3. Here, we provide an interpretation of each term:

- The causal contribution $\Delta C(q_j^+; \mathbf{q}) > 0$ is the *positive* increment of information that a source state $\mathbf{q}$ provides about a future target state q_j^+ , resulting in a decrease in uncertainty about q_j^+ .
- The non-causal contribution $\Delta \mathcal{N}(q_j^+; \mathbf{q}) \leq 0$ is the *negative* increment of information obtained by observing $\mathbf{q}$, which leads to increased uncertainty about the future state q_j^+ .
- The causality leak $\Delta I_{\text{leak} \rightarrow j} \geq 0$ is the effect of *unobserved* variables that influence Q_j^+ but are not contained in $\mathbf{Q}$, i.e., the information of Q_j^+ that remains unexplained after having observed $\mathbf{Q}$.

To illustrate the distinction between $\Delta C(q_j^+; \mathbf{q})$ and $\Delta \mathcal{N}(q_j^+; \mathbf{q})$, consider the task of forecasting whether it will rain tomorrow, denoted by q_{rain}^+ , where $q_{\text{rain}}^+ = 1$ indicates rain and $q_{\text{rain}}^+ = 0$ indicates no rain. Suppose we use an atmospheric pressure sensor as the observable, Q_p , which can take on two values: low ($q_p = 0$) or high ($q_p = 1$). Within our causality framework, if observing low pressure ($q_p = 0$) provides a positive increment of information (i.e., reduces uncertainty) about the state $q_{\text{rain}}^+ = 1$, then $q_p = 0$ is considered causal and contributes to $\Delta C(q_{\text{rain}}^+; q_p)$. Conversely, if observing $q_p = 0$ yields a negative increment of information (i.e., increases uncertainty) about $q_{\text{rain}}^+ = 0$, then $q_p = 0$ is regarded as non-causal and contributes to $\Delta \mathcal{N}(q_{\text{rain}}^+; q_p)$.

The final step is to decompose $\Delta C(q_j^+; \mathbf{q})$ and $\Delta \mathcal{N}(q_j^+; \mathbf{q})$ into distinct types of interactions—namely, redundant, unique, and synergistic components³³:

$$\Delta C(q_j^+; \mathbf{q}) = \sum_{i \in \mathcal{P}} \Delta C_{i \rightarrow j}^R + \sum_{i=1}^N \Delta C_{i \rightarrow j}^U + \sum_{i \in \mathcal{P}} \Delta C_{i \rightarrow j}^S, \quad (2)$$

where $\Delta C_{i \rightarrow j}^R$, $\Delta C_{i \rightarrow j}^U$, and $\Delta C_{i \rightarrow j}^S$ are the redundant, unique, and synergistic causalities, respectively, from the source states $\mathbf{q}$ to target state q_j^+ . Unique causalities are associated with individual components of $\mathbf{q}$, whereas redundant and synergistic causalities arise from groups of components from $\mathbf{q}$. Consequently, the set $\mathcal{P}$ contains all combinations involving more than one variable. For instance, for $N=2$, Eq. (2) reduces to $\Delta C(q_j^+; \mathbf{q}) = \Delta C_{12 \rightarrow j}^R + \Delta C_{1 \rightarrow j}^U + \Delta C_{2 \rightarrow j}^U + \Delta C_{12 \rightarrow j}^S$. Similar to the causal

and non-causal contributions, we provide an interpretation of redundant, unique and synergistic causalities, with the mathematical details available in Methods and the Supplementary Note 1.3.

- Redundant causality from $\mathbf{q}_i = [q_{i_1}, q_{i_2}, \dots]$ to q_j^+ (denoted by $\Delta C_{i \rightarrow j}^R$) refers to the causal influence shared among all individual states in $\mathbf{q}_i$, where $\mathbf{q}_i$ is a subset of $\mathbf{q}$. Redundant causality occurs when observing any of the states q_{i_1}, q_{i_2} , etc., provides identical amount of information about the outcome q_j^+ . Consider again the example of rain forecasting: we might use two pressure sensors placed in close proximity, both offering redundant causal information—i.e., each is equally useful for predicting whether it will rain or not.
- Unique causality from q_i to q_j^+ (denoted by $\Delta C_{i \rightarrow j}^U$) is the causal influence from q_i that cannot be obtained from any other individual variable different from Q_i . This occurs when observing q_i yields more information about q_j^+ than observing all the states of any other individual variable. In the previous example, this will occur when one pressure sensor provides more information about the likelihood of rain than the other.
- Synergistic causality from $\mathbf{q}_i = [q_{i_1}, q_{i_2}, \dots]$ to q_j^+ (denoted by $\Delta C_{i \rightarrow j}^S$) refers to the causal influence that arises from the joint effect of the states $\mathbf{q}_i$. Synergistic causality occurs when more information about q_j^+ is gained by observing the collection of states $[q_{i_1}, q_{i_2}, \dots]$ simultaneously than by observing all the states of subsets of $\mathbf{q}_i$ individually. In the rain forecasting scenario, a synergy would take place when the information gained by measuring with two pressure sensors simultaneously is greater than the information provided by each sensor individually. An analogous decomposition (redundant, unique, and synergistic) applies to the non-causal term $\Delta \mathcal{N}(q_j^+; \mathbf{q})$. However, in this case, the information gain is negative—acquiring this knowledge actually increases uncertainty about future outcomes. Here, we focus on the causal contribution $\Delta C(q_j^+; \mathbf{q})$. The non-causal components also offer valuable insight, with further discussion and examples provided in the Supplementary Note 1.4.

By definition, causal contributions are assigned following the order: redundant $\rightarrow$ unique $\rightarrow$ synergistic. Redundant causalities are identified first. The next increments of information that are not redundant are attributed to unique causality. Finally, the remaining causal influence is classified as synergistic. This implies that when contributions are labeled as causal (information gain) or non-causal (information loss), the gain/loss is referred to the previously identified redundant, unique, or synergistic causality. For example, in a system with one redundant, one unique, and one synergistic contribution: redundant causality reflects a gain relative to the information in the target q_j^+ ; unique causality reflects a gain relative to the redundant component; and synergistic causality reflects a gain relative to the unique one. This process is illustrated in the diagrams in Fig. 1, where the causal and non-causal contributions between the source states q_1 and q_2 and the target state q_j^+ are decomposed into its redundant, unique, and

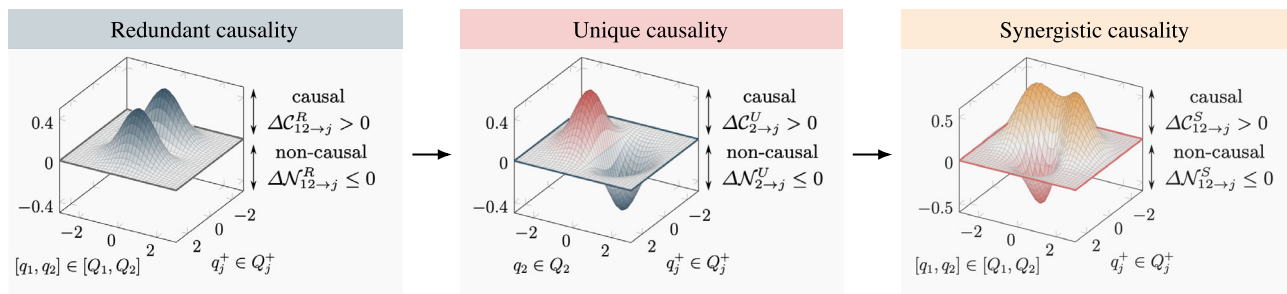


Fig. 1 | Decomposition of causality by states and interaction type. Diagram of the decomposition of causal dependencies between a vector of observed states $\mathbf{q} = [q_1, q_2]$ and a future target state q_j^+ into synergistic (S), unique (U), and redundant (R) components—shown in orange, red, and blue, respectively. Note that redundant and synergistic interactions depend on all three states, $[q_1, q_2, q_j^+]$, but the source $[q_1, q_2]$ is shown along a single axis for schematic simplicity. The causal

contributions are assigned following the order: redundant $\rightarrow$ unique $\rightarrow$ synergistic. As a result, causal states are color-coded according to the type of causality they contribute to, whereas non-causal states are shown using the color of the preceding causality level. For example, in the unique causality panel, causal states are shown in red, while non-causal states are shown in blue. This approach prevents the double counting of causal influences.

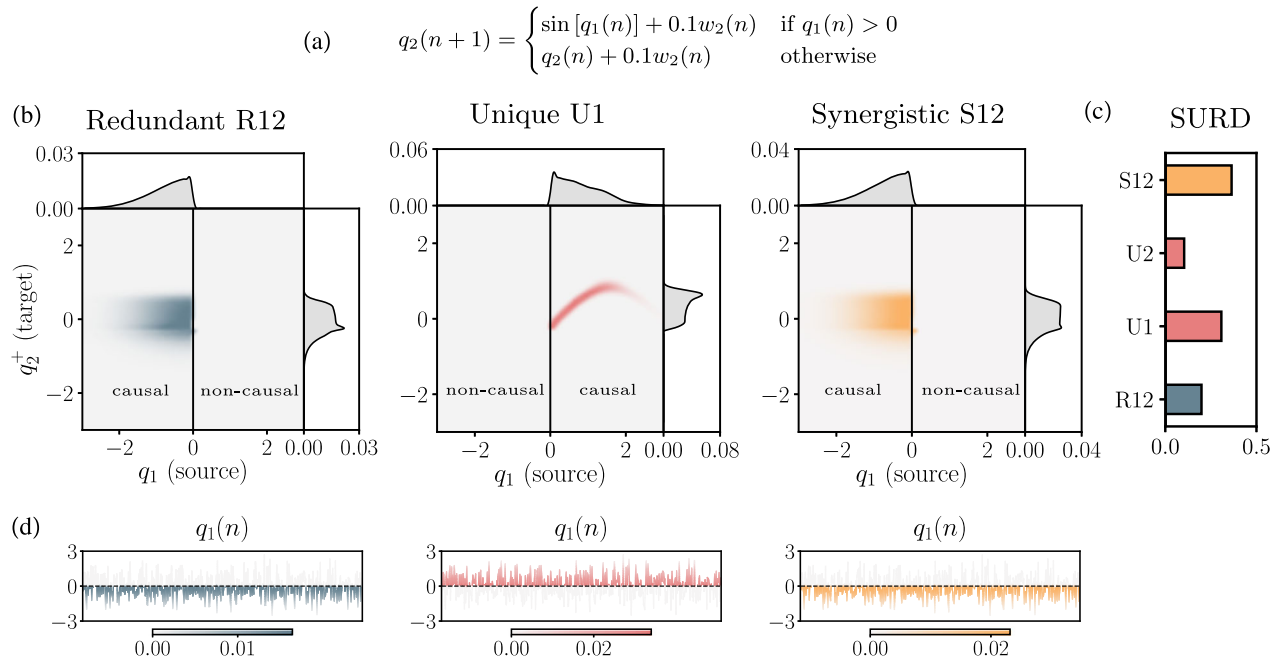


Fig. 2 | State-dependent decomposition of causality for the source-dependent case. **a** Equations governing the source-dependent system, where q_1 and q_2 represent the states of the variables Q_1 and Q_2 , respectively, and n denotes the time instant at which the variables are evaluated. Each variable is affected by the external noise terms w_1 and w_2 . **b** State-dependent redundant ($\Delta C_{12 \rightarrow 2}^R$), unique ($\Delta C_{12 \rightarrow 2}^U$), and synergistic ($\Delta C_{12 \rightarrow 2}^S$) causal contributions in blue, red, and orange, respectively. The causal maps for redundant and synergistic components show averages over all states of q_2 . **c** SURD causalities to the target variable Q_2^+ . The notation follows: U1 denotes

the unique SURD causality from Q_1 to Q_2^+ , i.e., $\Delta I_{1 \rightarrow 2}^U$, with analogous definitions for other components (i.e., R and S). All SURD and state-dependent causal contributions are normalized by the mutual information $I(Q_2^+; Q_1, Q_2)$. **d** Temporal evolution of $q_1(n)$, color-coded according to the corresponding instantaneous state-dependent causal contribution. The color maps represent the contributions to $\Delta C_{12 \rightarrow 2}^R$, $\Delta C_{12 \rightarrow 2}^U$, and $\Delta C_{12 \rightarrow 2}^S$, respectively, and their color intensities match those used in (b).

synergistic components. This approach ensures that causal contributions are not double-counted if they have already been accounted for in the previous type of interaction.

Our approach also enables the representation of causality as a function of time. The instantaneous value of causality at each time step is obtained by identifying the current state of the system and extracting the corresponding state-dependent value of $\Delta C_{i \rightarrow j}^R$, $\Delta C_{i \rightarrow j}^U$, and $\Delta C_{i \rightarrow j}^S$. As a result, it becomes possible to visualize the dynamic evolution of causal interactions, capturing temporal reversals and highlighting periods dominated by different types of causal contributions.

Finally, we define the cumulative causality over all possible states, after accounting for the uncertainty introduced by non-causal contributions, as $\Delta I_{i \rightarrow j}^\alpha = \sum_{q_j^+ \in Q_j^+} \sum_{q \in Q} [\Delta C_{i \rightarrow j}^\alpha + \Delta N_{i \rightarrow j}^\alpha]$, for $\alpha \in [R, U, S]$. These cumulative increments of information directly yield the causalities from SURD proposed in ref. 53. This is a key feature of our method, as SURD causality has been shown to outperform other definitions of causality across a wide range of benchmark cases.

Validation

We demonstrate our causal inference approach using validation cases representative of two common types of causal relationships: source-dependent and target-dependent causality. Each system consists of two source variables, Q_1 and Q_2 , and a target variable, Q_2^+ , designed such that the direction of causality varies with the intensity of either the source or the target. These benchmark cases are constructed so that the direction of causality for each state is known a priori, allowing us to assess the accuracy and reliability of the method. Importantly, the cases are also designed to exhibit qualitatively similar SURD causalities—i.e., comparable cumulative increments $\Delta I_{i \rightarrow 2}^\alpha$ —while the underlying causal flow reveals an entirely opposite structure as a function of the system state, a distinction that can only be revealed by a causality-by-state decomposition. Additional validation cases are presented in the Supplementary Note 2.

Source-dependent causality. We consider the two-variable system, $[Q_1, Q_2]$, in which the direction of causality toward the future state Q_2^+ depends on the value of the source variable Q_1 . Specifically, $Q_1 \rightarrow Q_2^+$ when $q_1 > 0$, and $Q_2 \rightarrow Q_2^+$ when $q_1 \leq 0$. The equation governing the system is provided in Fig. 2. The variable Q_1 is sampled from a normal distribution with zero mean and unit variance. The variable Q_2 is influenced by an external noise term, W_2 , drawn from a normal distribution with mean -2 and unit standard deviation. This noise term is not included in the vector of observables and serves to mimic unobserved perturbations commonly encountered in real-world systems.

The state-dependent redundant, unique, and synergistic causalities for the target variable Q_2^+ are shown in Fig. 2. The figure also includes the time evolution of the causalities, which is obtained by mapping the state-aware causality to the corresponding time instances.

For states with $q_1 > 0$, the results clearly show a nonzero contribution from unique causality ($\Delta C_{1 \rightarrow 2}^U$). In this regime, Q_1 acts as the sole driver of Q_2^+ , providing unique information that is not contained in Q_2 . For states with $q_1 < 0$, both redundant ($\Delta C_{12 \rightarrow 2}^R$) and synergistic ($\Delta C_{12 \rightarrow 2}^S$) causalities coexist. Redundancy arises because, in the region $q_1 < 0$, the information contained in q_1 is already provided (i.e., rendered redundant) by q_2 , which fully determines the outcome. This leads to duplicated information about Q_2^+ from both Q_1 and Q_2 . Synergistic causality emerges because two pieces of information are simultaneously needed to predict Q_2^+ , i.e., the value of Q_1 , to assess the condition $q_1 < 0$, and the value of Q_2 , to determine the outcome. Thus, Q_1 and Q_2 jointly provide information about Q_2^+ that neither can supply independently. Figure 2 shows the cumulative causal contributions to Q_2^+ based on the SURD causalities ($\Delta I_{i \rightarrow 2}^\alpha$). These results, which are consistent with the discussion above, will serve as a reference for the next benchmark case, where we demonstrate that the underlying state-dependent causal structures follow opposite trends despite exhibiting similar SURD causalities.

Target-dependent causality. The second benchmark case involves a system in which the direction of causality depends on the state of the

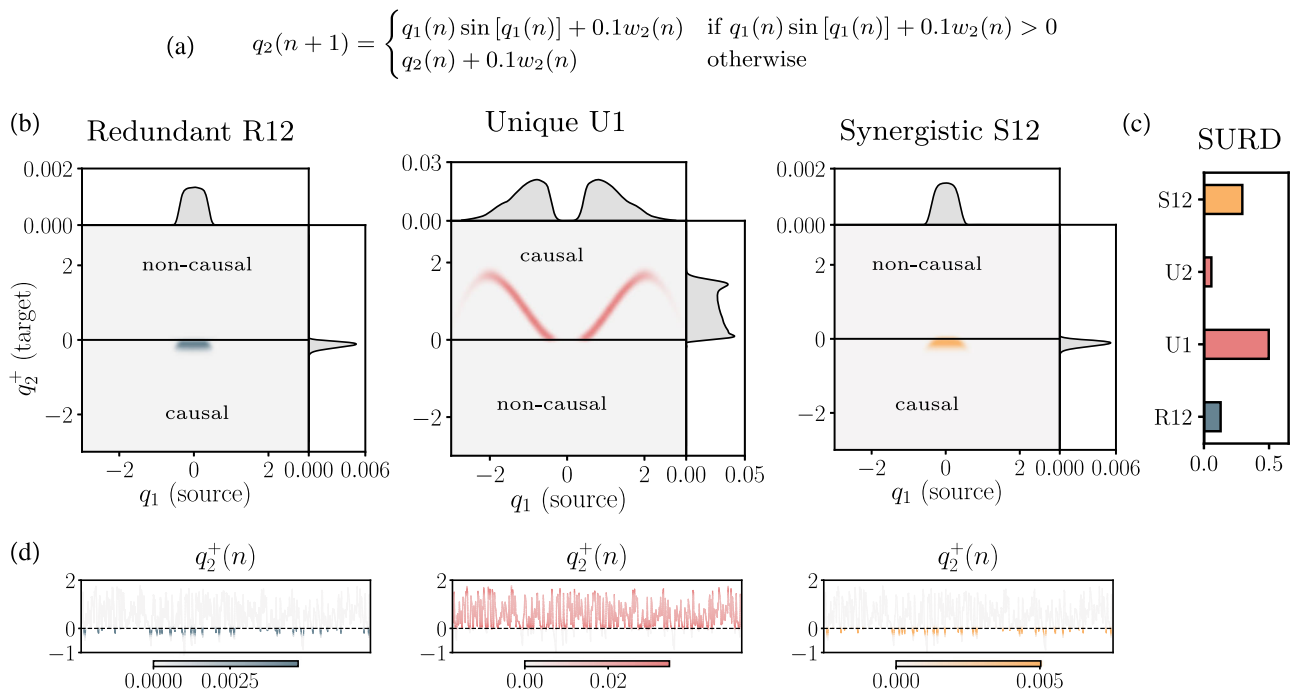


Fig. 3 | State-dependent decomposition of causality for the target-dependent case. **a** Equations governing the target-dependent system, where q_1 and q_2 represent the states of the variables Q_1 and Q_2 , respectively, and n denotes the time instant at which the variables are evaluated. Each variable is affected by the external noise terms w_1 and w_2 . Here, the target-dependent condition is given by the nonlinear function $f(q_1) = q_1 \sin(q_1) + 0.1w_2$, which also determines the future value of the target variable when $f(q_1) > 0$. **b** State-dependent redundant ($\Delta C_{12 \rightarrow 2}^R$), unique ($\Delta C_{12 \rightarrow 2}^U$), and synergistic ($\Delta C_{12 \rightarrow 2}^S$) causal contributions in blue, red, and orange, respectively. The causal maps for redundant and synergistic causalities are shown for

the q_2 that maximizes the total sum of individual contributions. **c** SURD causalities to the target variable Q_2^+ . The notation follows: U1 denotes the unique SURD causality from Q_1 to Q_2^+ , i.e., $\Delta I_{1 \rightarrow 2}^U$, with analogous definitions for other components (i.e., R and S). All SURD and state-dependent causal contributions are normalized by the mutual information $I(Q_2^+; Q_1, Q_2)$. **d** Temporal evolution of $q_2^+(n)$, color-coded according to the corresponding instantaneous state-dependent causal contribution. The color maps represent the contributions to $\Delta C_{12 \rightarrow 2}^R$, $\Delta C_{12 \rightarrow 2}^U$, and $\Delta C_{12 \rightarrow 2}^S$, respectively, and their color intensities match those used in (b).

target variable. As shown in Fig. 3, the system again consists of two variables, $[Q_1, Q_2]$, such that $f(Q_1) \rightarrow Q_2^+$ when $f(q_1) > 0$, and $Q_2 \rightarrow Q_2^+$ when $f(q_1) \leq 0$. Here, $f(\cdot)$ denotes a nonlinear function $f(q_1) = q_1 \sin(q_1) + 0.1w_2$, which maps the past of the source variable to the future of the target. The variable Q_1 is sampled from a normal distribution with zero mean and unit variance, while Q_2 is influenced by an external noise term, W_2 , drawn from a normal distribution with unit mean and unit standard deviation.

The state-dependent redundant, unique, and synergistic causalities for the target variable Q_2^+ are shown in Fig. 3, along with the corresponding time evolution of the causalities. The analysis clearly shows that the unique causality, $\Delta C_{1 \rightarrow 2}^U$, is primarily determined by the state of the target variable rather than the source. In particular, positive target states ($q_2^+ > 0$) exhibit the strongest unique influence from Q_1 , consistent with the fact that Q_1 acts as the sole driver of Q_2^+ in this regime. In contrast, negative values of q_2^+ give rise to both synergistic ($\Delta C_{12 \rightarrow 2}^S$) and redundant ($\Delta C_{12 \rightarrow 2}^R$) causalities, as both Q_1 and Q_2 are required to predict Q_2^+ when $f(q_1) \leq 0$. This behavior mirrors the reasoning observed in the source-dependent benchmark case.

It is worth noting that the cumulative SURD causality shown in Fig. 3 resembles that of the source-dependent case discussed earlier, despite the fundamentally different interactions between the system variables. This illustrates a clear case where a state-aware decomposition of causality is essential to uncover causal pathways that would be overlooked by a state-averaged inference approach.

Application to flow interactions in wall turbulence

We apply our state-dependent causal inference method to identify the most causal flow states within a turbulent boundary layer, i.e., the chaotic fluid motion within a thin region adjacent to solid surfaces. Turbulent boundary layers are responsible for nearly 50% of the aerodynamic drag on modern

airliners and play a critical role in the lower atmosphere, particularly within the first hundred meters, where they influence broader meteorological phenomena⁵⁷. Understanding and modeling the interactions among turbulent motions of different scales within these layers remains a major challenge due to strong nonlinearities and high dimensionality of the system. Prior studies have documented the impact of large-scale motions in the outer layer on smaller-scale motions near the wall^{53,57–65}, supporting the notion of top-down causality (a.k.a. Townsend's outer-layer similarity hypothesis⁶⁶). In this work, we focus on identifying the specific flow states that contribute to the causal influence from the outer layer flow (far from the wall) to the inner layer flow (near the wall).

We utilize data from a high-fidelity numerical simulation of turbulent flow over a flat plate with zero mean-pressure gradient, which provides a realistic representation of turbulent boundary layers encountered in engineering applications⁶⁷. Figure 4 shows an instantaneous visualization of the streamwise velocity. The flow conditions are characterized by the friction Reynolds number, $Re_\tau = u_\tau \delta / \nu$, which ranges from around 300 (inflow) to 700 (outflow), where δ is the boundary-layer thickness, ν is the kinematic viscosity, and u_τ is the friction velocity at the wall. The time signals analyzed here correspond to the streamwise velocity measured at two wall-normal locations, representing the inner and outer layers of the boundary layer. These velocity signals, denoted as $u_i(t)$ and $u_o(t)$, are located at wall-normal distances of $y_i = 4\nu/u_\tau$ and $y_o = 0.3\delta$, respectively. Figure 4 presents a sample of these velocity signals at $Re_\tau = 500$.

The objective is to identify the states that contribute most significantly to the causal influence of the outer-layer velocity $u_o(t)$ on the future inner-layer velocity $u_i^+(t) = u_i(t + \Delta T)$. The time lag ΔT for the causal evaluation is selected to maximize the unique causality from $u_o(t)$ to $u_i^+(t)$, resulting in an optimal delay of approximately $\Delta T \approx 60\nu/u_\tau^2$. The causal influence in the reverse direction, from $u_i(t)$ to $u_o(t + \Delta T)$, is discussed in

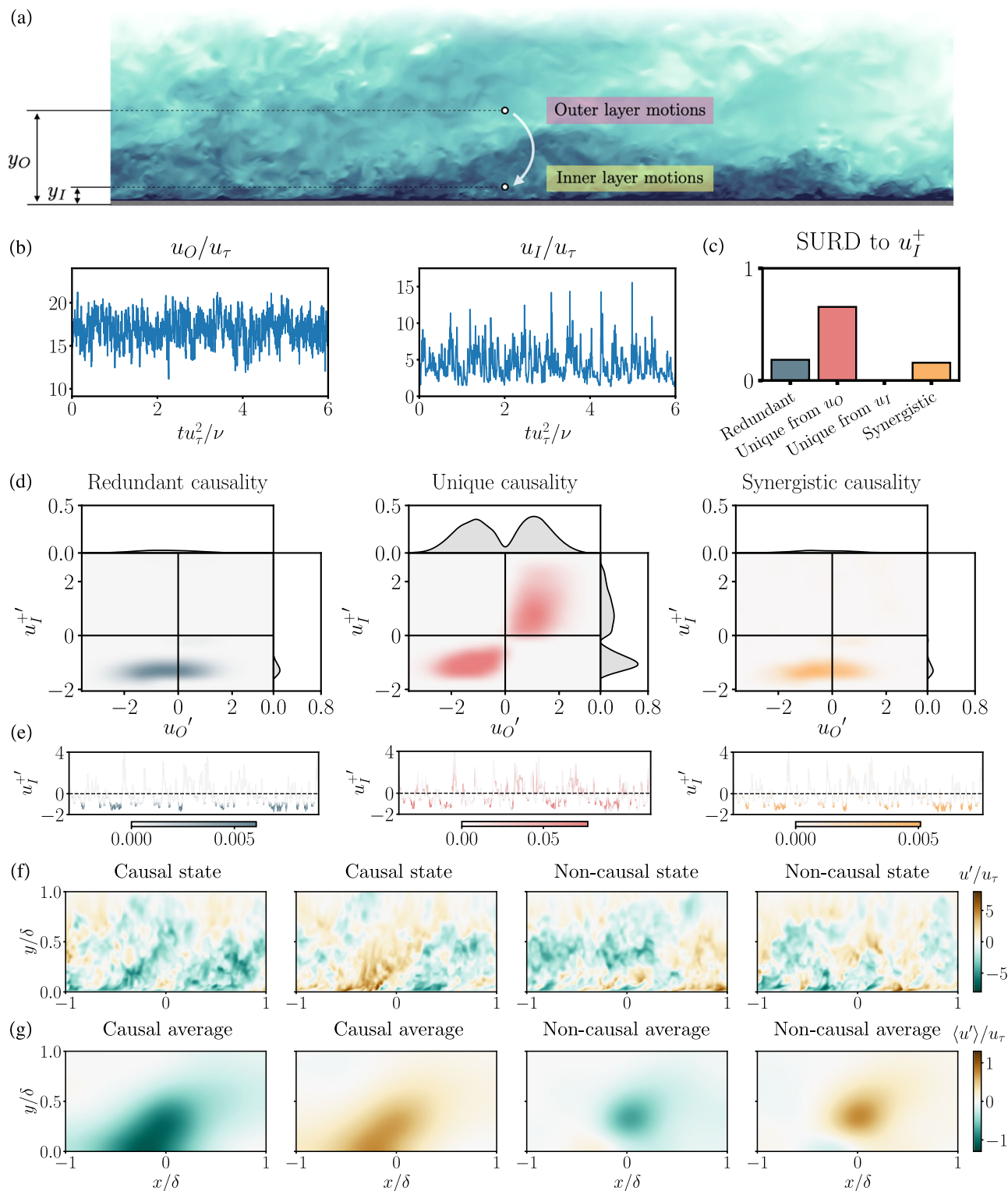


Fig. 4 | Causality between inner and outer flow motions in a turbulent boundary layer. **a** Instantaneous visualization of the streamwise velocity field. **b** Temporal evolution of the streamwise velocity at two fixed wall-normal locations: the outer layer ($y_O = 0.3\delta$) and the inner layer ($y_I = 4\nu/u_\tau$). **c** Redundant ($\Delta I_{O \rightarrow I}^R$), unique ($\Delta I_{O \rightarrow I}^U$), and synergistic ($\Delta I_{O \rightarrow I}^S$) SURD causalities to the future inner-layer velocity $u_I^+ = u_I(t + \Delta T)$. The time lag ΔT is chosen to maximize cross-induced unique causality. **d** State-dependent redundant ($\Delta C_{O \rightarrow I}^R$), unique ($\Delta C_{O \rightarrow I}^U$), and synergistic ($\Delta C_{O \rightarrow I}^S$) causalities as functions of the instantaneous streamwise velocities u_O (outer layer) and u_I^+ (inner layer). Note that for some of the panels, the color

scale has been intentionally saturated to enhance visual contrast; however, the causalities are of small magnitude, as evidenced by the horizontal and vertical projections. **e** Time series of u_I^+ color-coded by the dominant state-dependent causal component at each time step. The color maps represent the contributions to $\Delta C_{O \rightarrow I}^R$, $\Delta C_{O \rightarrow I}^U$, and $\Delta C_{O \rightarrow I}^S$, respectively, and their color intensities match those used in (d). **f** Instantaneous and **g** conditionally averaged visualizations of flow fields corresponding to the causal and non-causal states, respectively. The color maps show the fluctuation of the (f) instantaneous streamwise velocity u' and (g) conditionally averaged streamwise velocity ($\langle u' \rangle$), each normalized by the friction velocity u_τ .

the Methods section. We adopt the notation $(\cdot)'$ to indicate standardization of the signals by their respective mean and standard deviation. Thus, negative values of u_O' represent below-average deviations of u_O , while positive values correspond to above-average deviations, with the same convention applied to u_I' .

We begin by analyzing the SURD causalities from $[u_I, u_O]$ to u_I^+ , as shown in Fig. 4. The redundant, unique, and synergistic components are denoted by $\Delta I_{O \rightarrow I}^R$, $\Delta I_{O \rightarrow I}^U$, and $\Delta I_{O \rightarrow I}^S$, respectively, where the subscripts indicate contributions from the outer-layer (O) and inner-layer (I) velocities. The results demonstrate that the unique causality from u_O to u_I^+ dominates, followed by the synergistic and redundant components. In contrast, the unique causality from u_I to u_I^+ is negligible at the current time lag ΔT , since this influence occurs on timescales shorter than ΔT^{68} .

We now examine the state-dependent decomposition of causality, presented in Fig. 4, which also illustrates the temporal evolution of the individual causal components. The dominant form of causality is the unique component, which exceeds the redundant and synergistic components by more than an order of magnitude. The small redundant and synergistic causalities are concentrated in regions where $u_I^+ < 0$. This indicates that, during low-speed motions in the inner layer (i.e., velocities below the mean flow), part of the causal influence arises from overlapping information simultaneously conveyed by the outer- and inner-layer histories, while another portion is jointly determined by concurrent fluctuations in both layers.

The unique causal contribution is predominantly distributed within the first ($u_O' > 0, u_I^+ > 0$) and third ($u_O' < 0, u_I^+ < 0$) quadrants of the $u_O'-u_I^+$ space. These quadrants represent states where outer- and inner-layer motions concurrently exceed or fall below the mean velocity. Moreover, the region of causal states $\Delta C_{O \rightarrow I}^U$ in the first quadrant is wider than the third, consistent with previous studies⁶¹ showing that large positive fluctuations in the outer layer exert a stronger influence on small-scale inner-layer structures compared to negative fluctuations. This observation supports the well-known modulation effect, whereby high-speed outer-layer events leave a stronger causal imprint on the inner-layer dynamics⁶¹. In contrast, the second and fourth quadrants do not contain causal contributions. Overall, these findings indicate that high-speed streaks propagating towards the wall act as dominant drivers of unique causality. Conversely, low-speed streaks are predominantly associated with redundant or synergistic causal interactions.

We conclude this section by examining the flow structures associated with causal and non-causal states. To this end, we analyze instantaneous realizations of the streamwise velocity field, $u(x, z, t)$, within a spatial region of size 2δ , conditioned on states of $u_O(t)$ and $u_I^+(t)$ corresponding to the four quadrants of the unique causality map $\Delta C_{O \rightarrow I}^U$ discussed previously. Figure 4 shows example snapshots of $u(x, z, t)$ representative of each quadrant. To highlight characteristic flow patterns for each quadrant, we compute ensemble-averaged velocity fields, $\langle u(x, z, t) \rangle$, over all instances belonging to the respective quadrant. For instance, the flow field in the first quadrant is averaged over all time instants for which $u_O' > 0$ and $u_I^+ > 0$, and similarly for the other quadrants. The results (bottom of Fig. 4) reveal that causal states correspond to coherent velocity structures attached to the wall, extending beyond $y = 0.3\delta$, the wall-normal location of u_O . These findings indicate that causal interactions are associated with coherent flow structures—such as high- or low-speed streaks—that span across the inner and outer layers. Furthermore, the shape of these structures is in agreement with other averaged flow structures found in wall-bounded turbulence⁶⁹. In contrast, the non-causal states (second and fourth quadrant) do not exhibit such structural coherence, with velocity fields appearing fragmented and detached from the wall. This fragmentation shows a breakdown of spatial coherence and diminished influence from the outer to the inner layer.

Application to Walker circulation

We apply our state-dependent causal inference method to analyze the Walker circulation in the tropical Pacific—a key atmospheric process with a well-established physical mechanism⁷⁰, and a subject of previous causal

investigations^{4,16}. However, a state-aware analysis that accounts for redundant and unique causalities has not yet been performed. The Walker circulation is a large-scale atmospheric circulation pattern in the tropical Pacific, driven by the temperature gradient between the warm waters of the Western Pacific and the cooler Eastern Pacific. Under normal conditions, strong trade winds transport warm surface waters westward, leading to moist air rising over the Western Pacific due to enhanced convection. This air ascends, moves eastward aloft, cools, and descends over the Eastern Pacific. The circulation is completed as the cooler, drier air flows westward near the surface, forming a closed loop across the tropical Pacific. Figure 5 illustrates this circulation pattern, which governs critical atmospheric and oceanic interactions in the Pacific basin and plays a central role in driving global weather patterns and climate variability.

We use data consisting of regional averages of surface pressure anomalies in the West Pacific (WPAC) and surface air temperature anomalies in the Central Pacific (CPAC) and East Pacific (EPAC) regions for the period 1948–2012, comprising 780 months¹⁶. The regions used for these averages are illustrated as shaded boxes in Fig. 5. Here, we focus on understanding what particular states of each variable are responsible for the Walker circulation phenomenon from the East to West Pacific close to the sea surface. To achieve this, we analyze the most relevant total redundant and cross-induced unique causalities to CPAC and WPAC involving at most two variables, which are shown in Fig. 5. The number of samples in the dataset was insufficient to reliably estimate synergistic causalities, which are therefore not included in the analysis.

The decomposition for these causalities is shown in Fig. 5, as a function of the values of the source and target variables. First, we focus on the redundant and cross-induced unique causalities to WPAC⁺. In this case, unique causalities dominate and redundant causalities are small. The small redundant causality shows causal states for negative values of CPAC and WPAC⁺, whereas the higher causal states for the unique causality occur when both CPAC and WPAC⁺ are either above or below their respective means. Notably, states above the mean exhibit stronger causal influence compared to those below⁷¹. For the rest of the cases, i.e., where CPAC and WPAC⁺ have opposite signs, the state-dependent unique causality is zero, which indicates that causality is already explained by the redundant components. These results are consistent with the observation that positive CPAC temperature anomalies (associated with El Niño conditions) have a more pronounced impact on Western Pacific surface pressure than negative anomalies (associated with La Niña conditions)^{71,72}.

For CPAC⁺, the most significant contributions are the redundant [CPAC, EPAC] $\rightarrow$ CPAC⁺ and unique EPAC $\rightarrow$ CPAC⁺ causalities. We find that the values of EPAC and CPAC⁺ contributing most significantly to EPAC $\rightarrow$ CPAC⁺ come from the extreme values of EPAC and CPAC⁺, far from the mean. In contrast, EPAC values near the mean are identified as non-causal, as their causality to CPAC⁺ is already explained by the redundant component [CPAC, EPAC] $\rightarrow$ CPAC⁺, with causal states concentrated around positive values of EPAC and CPAC⁺. This result supports previous observations⁷³, where it was reported that positive temperature anomalies in the Central Pacific can originate from positive temperature anomalies in the Eastern Pacific⁷⁴.

Overall, our findings indicate that causal states for sea surface temperatures in the Central Pacific (CPAC) and Eastern Pacific (EPAC), as well as surface pressure in the Western Pacific (WPAC), predominantly occur when all three variables are either simultaneously above (EPAC > 0, CPAC > 0, WPAC > 0) or below (EPAC < 0, CPAC < 0, WPAC < 0) their respective means. In contrast, non-causal states emerge when there is a mismatch in trends between CPAC and WPAC, while EPAC remains near its mean. Specifically, non-causal interactions are observed when CPAC is below the mean and WPAC is above (CPAC < 0, WPAC > 0), or vice versa (CPAC > 0, WPAC < 0).

Discussion

We have introduced an information-theoretic causal inference method that decomposes causality according to both the system state and the interaction

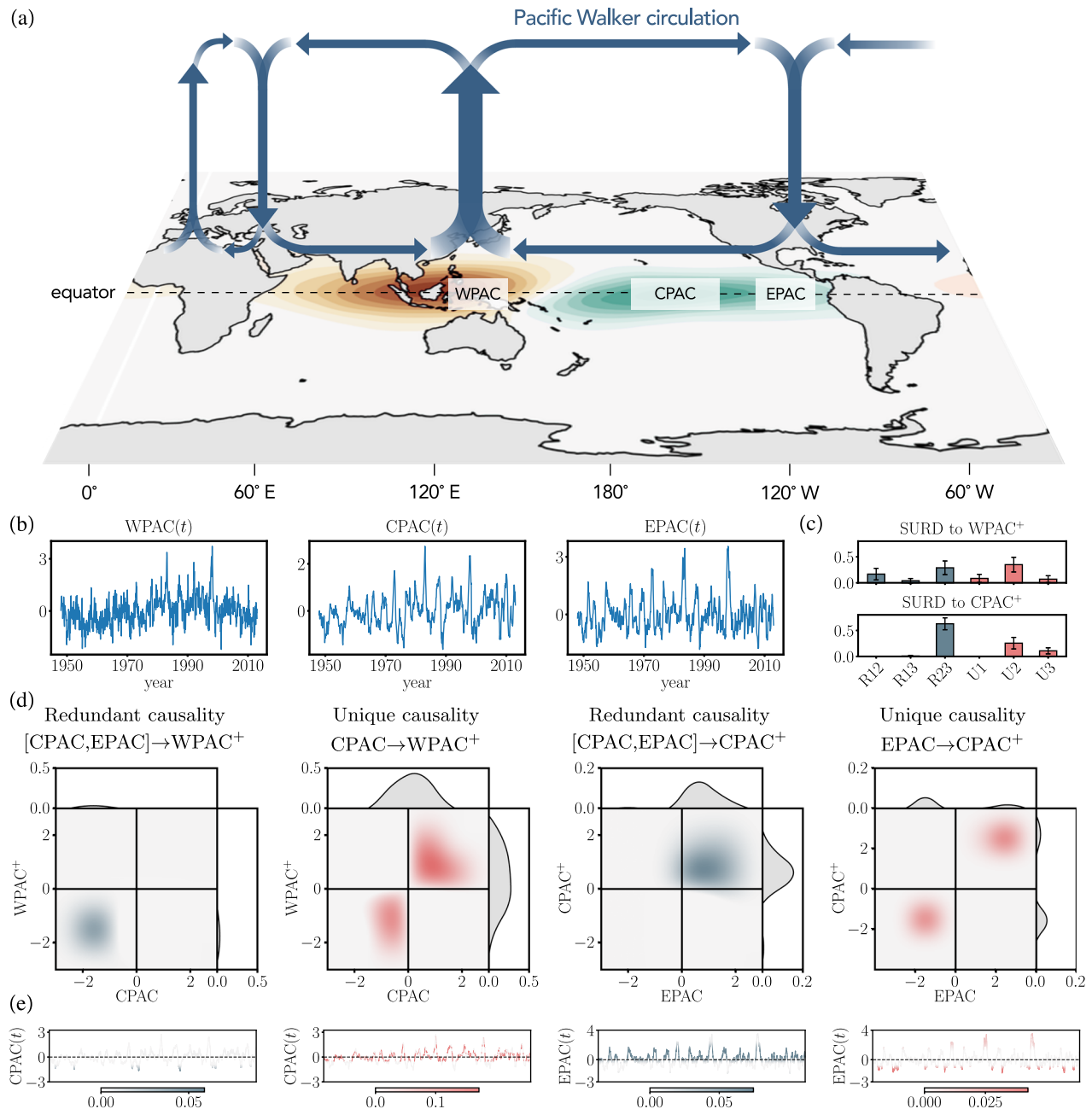


Fig. 5 | Causality in the Pacific Walker circulation. **a** Schematic representation of the Pacific Walker circulation mechanism, showing the regions where regional averages were computed for the period 1948–2012 ($T = 780$ months): surface pressure anomalies in the West Pacific (WPAC) and surface air temperature anomalies in the Central Pacific (CPAC) and East Pacific (EPAC). The shaded boxes indicate the specific regions of interest. The schematic diagram was adapted from ref. 91 and the data was taken from Watanabe et al.⁹², which illustrates how local sea surface temperature (SST) warming affects the strength of the Pacific Walker circulation, with positive (orange) values indicating strengthening and negative (green) values indicating weakening. **b** Time evolution of WPAC, CPAC, and EPAC. **c** Redundant (R) and unique (U) SURD causalities to $WPAC^+ = WPAC(t + \Delta T)$ and $CPAC^+ = CPAC(t + \Delta T)$. The variables WPAC, CPAC, and EPAC are denoted as 1, 2, and 3, respectively. For example, the label $R13 \rightarrow CPAC^+$ indicates the redundant causality from WPAC and EPAC to $CPAC^+$. The causal analysis is performed with a

time lag of $\Delta T = 4$ months, which corresponds to the maximum cross-induced causality. Error bars on the SURD causalities represent the standard deviation across 50 bootstrap resamplings. Statistical significance was assessed via 1000 random permutations, yielding a p -value of zero for both redundant and unique components. **d** Decomposition of state-dependent causalities: redundant $[CPAC, EPAC] \rightarrow WPAC^+$, unique $CPAC \rightarrow WPAC^+$, redundant $[CPAC, EPAC] \rightarrow CPAC^+$, and unique $EPAC \rightarrow CPAC^+$ as a function of the source and target variable states. Note that in the left-most panel, the color scale has been intentionally saturated to enhance visual contrast; however, the actual causalities are of small magnitude, as indicated by the horizontal and vertical projections. **e** Temporal evolution of $CPAC(t)$ and $EPAC(t)$ color-coded according to the state-dependent causal contribution at that instant. The color maps represent the contributions to each of the state-dependent causalities in (d), with color intensities consistent with those used in that panel.

type—namely, synergistic, unique, and redundant. This dual decomposition allows for the identification of causal system configurations, distinguishing them from non-causal states. The approach further enables visualization of causality over time, facilitating the detection of causal pathways. A key feature of the method is that summing the state-dependent causal and non-

causal contributions recovers the SURD causality⁵³, which has previously been shown to outperform existing methods for causal inference across diverse scenarios.

Two benchmark cases were designed to represent common causal pathways: source-dependent and target-dependent causality. In both systems,

the direction and strength of causal influence vary with the state of the interacting variables. Although the resulting SURD causalities for the two systems appear similar, their underlying causal flows are fundamentally different—a distinction clearly captured by our state-dependent causal inference method (Fig. 2). We have also compared our method to a state-dependent variant of CTE, detailed in Methods. Although CTE enables a state-wise evaluation of causality, it fails to detect source-dependent and target-dependent causal structures. This limitation arises because CTE aggregates all causal contributions into a single measure, without distinguishing between synergistic, unique, and redundant components. In general, we anticipate that existing causal inference methods—such as Granger causality, PCMC, or CCM—will likewise be unable to differentiate between source- and target-dependent causal pathways, since they rely on average causal strength and do not disentangle redundant and synergistic interactions.

We have applied our method to two real-world systems that had not previously been examined through the lens of state-level causality: scale interactions in a turbulent boundary layer and the Walker circulation in climate dynamics. In the turbulent boundary layer, our analysis revealed causal flow structures from the outer to the inner layer, demonstrating that high-speed streaks modulate near-wall motions in a state-specific manner (Fig. 4). In the Walker circulation case, we found that causal interactions between sea surface temperatures and surface pressure anomalies in the Eastern, Central, and Western Pacific emerge only when these variables deviate coherently from their mean values—highlighting the asymmetry between El Niño and La Niña regimes (Fig. 5). These detailed causal relationships are not captured by traditional causal inference methods that rely on state-aggregated measures.

Finally, it is informative to compare our method with other approaches proposed in the literature for analyzing the impact of source states on a target variable. Among these methods are local TE³⁴, specific TE⁴⁴, local GC³², and the time-varying Liang–Kleeman (TvLK) information flow⁵⁰. It is important to note that neither local TE nor specific TE are explicitly designed for causal inference—as discussed by their respective authors—but rather aim to quantify local information dynamics. Technical details of each approach are provided in the Supplementary Note 4, and results for the two benchmark systems shown in Fig. 2 are discussed in “Methods”. Here, we offer a brief summary. Local TE decomposes information transfer across both source and target states, yielding informative (positive) and misinformative (negative) contributions, while local GC can be viewed as a variation of local TE tailored for systems with linear relationships. Specific TE provides a decomposition based solely on source states, which has the benefit of yielding only positive contributions. Finally, TvLK relies on an estimation of the original Liang–Kleeman information flow formulation, employing a square-root Kalman filter to estimate covariance matrices over time⁵⁰. Although these approaches have proven useful for analyzing dynamical systems, they are not well suited to the problems considered here—namely, source-dependent and target-dependent systems—as their outcomes diverge from the intuitive causal pathways expected in such cases (see “Methods”). These limitations may be due to assumptions of linearity, difficulties in selecting an appropriate temporal scale, and the inability to distinguish redundant, unique, and synergistic contributions.

Overall, by providing a state-dependent, interaction-aware decomposition of causality, our method contributes a new perspective to the study of causal inference in complex systems. It offers a means to disentangle pathways for causal relationships and to identify specific conditions under which variables interact synergistically or redundantly. We hope that this approach can complement existing methods and help deepen our understanding of multiscale, nonlinear systems in fields such as climate science, fluid mechanics, neuroscience, and beyond.

Methods

Assumptions for causal discovery

The causal inference method proposed in this study follows an observational (i.e., non-interventional) approach within a probabilistic framework, inferring causal relationships from the transition probabilities between

system states under the assumption that explicit interventions are not necessary. Specifically, we employ the notion of causality proposed in ref. 53, in which causal influence is quantified through the informational gain associated with observing individual or groups of variables. The state-dependent causalities identified by our framework are consistent with the total causal contributions obtained from the SURD formulation⁵³.

We assume the causal Markov condition, which states that all relevant probabilistic information about a variable is contained in its direct causes (i.e., parents), rather than in indirect dependencies. This condition ensures that the inferred causal structure accurately reflects the true generative mechanism of the system. In addition, the method assumes faithfulness, i.e., observed statistical dependencies (or independencies) arise from the underlying causal structure itself, rather than from pathological parameter configurations or coincidences.

We further impose the principle of forward-in-time information propagation²⁹, which prohibits backward-in-time causation. To account for latent influences, our formulation incorporates the concept of a causality leak^{29,53}, which quantifies the portion of causal influence that remains unexplained by the observed variables. This allows us to relax the standard assumption of causal sufficiency, which traditionally requires all common causes to be explicitly measured.

The method is also *model-free*, requiring no prior knowledge of the governing equations or dynamics of the system. This generality makes the approach applicable to a wide variety of multivariate systems, whether deterministic or stochastic, and irrespective of whether the dependencies are linear or nonlinear. However, the analysis assumes that statistical properties of the input time series remain invariant or change slowly over time. This assumption enables the estimation of slowly varying non-stationary dynamics, provided there is sufficient statistical support within periods where time signals can be considered statistically stationary.

Finally, it is important to stress that our framework does not claim to provide an absolute or universal definition of causality in a strict mathematical sense. Rather, it offers a motivated definition, grounded in information theory, that is tailored to the analysis of complex multivariate systems. The goal is to deliver interpretable insights into system dynamics, rather than to prescribe a unique notion of causality. In this respect, our approach is aligned with other observational and probabilistic causal inference frameworks, which rely on tractable assumptions such as the discussed in this section.

Decomposition of redundant, unique, and synergistic causalities by states

To perform the decomposition proposed in Eq. (2), we start with the SURD approach, which is based on a decomposition of mutual information^{27,75,76}. The mutual information between a target variable Q_j^+ and a vector of source variables $\mathbf{Q}$ is given by:

$$I(Q_j^+; \mathbf{Q}) = \sum_{q_j^+ \in Q_j^+} \sum_{\mathbf{q} \in \mathbf{Q}} p(q_j^+, \mathbf{q}) \log_2 \left(\frac{p(q_j^+, \mathbf{q})}{p(q_j^+)p(\mathbf{q})} \right) \geq 0, \quad (3)$$

where q_j^+ and $\mathbf{q}$ denote the states of the target Q_j^+ and source $\mathbf{Q}$ variables, respectively. Mutual information quantifies how far the true joint distribution $p(q_j^+, \mathbf{q})$ deviates from the product $p(q_j^+)p(\mathbf{q})$, which would hold only if Q_j^+ and $\mathbf{Q}$ were statistically independent. When there is a degree of dependence between Q_j^+ and $\mathbf{Q}$, the joint density cannot be factorized in this way, and the difference between the two distributions reflects how much knowledge of the source variables $\mathbf{Q}$ reduces the uncertainty about Q_j^+ . The amount of uncertainty in Q_j^+ that cannot be accounted for by $\mathbf{Q}$ is known as the causality leak. In information-theoretic terms, this residual uncertainty is precisely the conditional Shannon entropy $H(Q_j^+ | \mathbf{Q})$ ^{27,27}:

$$H(Q_j^+ | \mathbf{Q}) = \sum_{q_j^+ \in Q_j^+} \sum_{\mathbf{q} \in \mathbf{Q}} -p(q_j^+, \mathbf{q}) \log_2 [p(q_j^+ | \mathbf{q})] \geq 0. \quad (4)$$

We decompose the mutual information into its redundant, unique, and synergistic components for each of the states of the target variable q_j^+ using the *specific* mutual information $\tilde{I}(Q_j^+ = q_j^+; \mathbf{Q})$, i.e., the mutual information between a fixed target state q_j^+ and all the source states in $\mathbf{Q}$ ^{53,77}:

$$\tilde{I}(q_j^+; \mathbf{Q}) = \sum_{\mathbf{q} \in \mathbf{Q}} \frac{p(q_j^+, \mathbf{q})}{p(q_j^+)} \log_2 \left(\frac{p(q_j^+, \mathbf{q})}{p(q_j^+)p(\mathbf{q})} \right) \geq 0. \quad (5)$$

For a given state q_j^+ of the target variable Q_j^+ , the specific causalities $\tilde{I}$ are computed for all the possible combinations of source variables. These components are organized in ascending order. The increments of information between each $\tilde{I}$, denoted by $\Delta\tilde{I}$, define the redundant $\Delta\tilde{I}_{i \rightarrow j}^R$, unique $\Delta\tilde{I}_{i \rightarrow j}^U$, and synergistic $\Delta\tilde{I}_{i \rightarrow j}^S$ causalities. Figure 6(left) shows an example for two source variables $\mathbf{Q} = [Q_1, Q_2]$ and a given state of the target variable q_j^+ . The example illustrates the specific mutual information $\tilde{I}_1, \tilde{I}_2, \tilde{I}_{12}$ and the corresponding redundant $\Delta\tilde{I}_{12 \rightarrow j}^R$ and unique $\Delta\tilde{I}_{2 \rightarrow j}^U$ increments. We denote by $\tilde{I}_{12 \rightarrow j}$ the specific mutual information between the future state q_j^+ of the target and the joint source $[Q_1, Q_2]$, i.e. $\tilde{I}_{12 \rightarrow j} = \tilde{I}(q_j^+; [Q_1, Q_2])$. For notational simplicity, when the target variable is clear from context we omit the index j and write $\tilde{I}_{12}$, using the same convention for $\tilde{I}_1, \tilde{I}_2, \Delta\tilde{I}_{12}^R$, and $\Delta\tilde{I}_{2 \rightarrow j}^U$.

The redundant, unique, and synergistic information increments are further decomposed into causal and non-causal contributions for each state pair $q_j^+ - \mathbf{q}$. These causal redundant, unique, and synergistic contributions are denoted as $\Delta C_{i \rightarrow j}^R, \Delta C_{i \rightarrow j}^U$, and $\Delta C_{i \rightarrow j}^S$, respectively. The general mathematical expressions for $\Delta C_{i \rightarrow j}^\alpha$ with $\alpha \in \{R, U, S\}$ are detailed in the Supplementary Note 1.3. Here, we provide the equations particularized for a system with two source variables $\mathbf{Q} = [Q_1, Q_2]$ and a target variable Q_j^+ , where $\tilde{I}_1 < \tilde{I}_2 < \tilde{I}_{12}$ is satisfied for all possible states $q_j^+ \in Q_j^+$:

$$\begin{aligned} \Delta C_{12 \rightarrow j}^R &= p(q_j^+, q_1, q_2) \max \left(0, \log_2 \frac{p(q_j^+ | q_1)}{p(q_j^+)} \right), \\ \Delta C_{2 \rightarrow j}^U &= \sum_{q_1 \in Q_1} p(q_j^+, q_1, q_2) \max \left(0, \log_2 \frac{p(q_j^+ | q_2)}{p(q_j^+ | q_1)} \right), \\ \Delta C_{12 \rightarrow j}^S &= p(q_j^+, q_1, q_2) \max \left(0, \log_2 \frac{p(q_j^+ | q_1, q_2)}{p(q_j^+ | q_2)} \right). \end{aligned} \quad (6)$$

Similar expressions can be derived for the non-causal components $\Delta \mathcal{N}_{i \rightarrow j}^\alpha$, which are detailed in the Supplementary Note 1.3. Figure 6 illustrates the interpretation of the unique causal and non-causal states, denoted by $\Delta C_{2 \rightarrow j}^U$ and $\Delta \mathcal{N}_{2 \rightarrow j}^U$, respectively. In this context, causal states (in red) indicate that q_2 provides more information about the q_j^+ than q_1 , i.e.,

$\log_2 p(q_j^+ | q_2) > \log_2 p(q_j^+ | q_1)$. Conversely, non-causal states (in blue) indicate that q_2 offers less or the same information about q_j^+ than q_1 , i.e., $\log_2 p(q_j^+ | q_2) < \log_2 p(q_j^+ | q_1)$.

The SURD causalities, $\Delta I_{i \rightarrow j}^\alpha$ where $\alpha \in \{R, U, S\}$, are recovered by summing the causal and non-causal contributions across all possible states:

$$\Delta I_{i \rightarrow j}^\alpha = \sum_{q_j^+ \in Q_j^+} \sum_{\mathbf{q} \in \mathbf{Q}} [\Delta C_{i \rightarrow j}^\alpha(q_j^+; \mathbf{q}) + \Delta \mathcal{N}_{i \rightarrow j}^\alpha(q_j^+; \mathbf{q})]. \quad (7)$$

Note that in SURD, the average causal effect of one variable on another is obtained by accounting for both causal and non-causal contributions. There are two main reasons for this construction. First, including both causal and non-causal contributions ensures consistency with the mutual information between variables, as illustrated in Fig. 1. This also guarantees consistency with the forward-in-time propagation of information condition from Eq. (1). The second reason is the need to define an average causal measure that discounts instances where knowledge of the source variable actually increases uncertainty about the target. In practice, if only positive (causal) contributions were considered, one might mistakenly deem a variable Q_1 causal to another variable Q^+ based on a small subset of states q_1 where the influence is strong. However, it could occur that, in the vast majority of instances, knowledge of q_1 does not increase, and may even decrease, certainty about Q^+ . Including non-causal (negative) contributions ensures that these misleading instances are properly accounted for, leading to an average measure that more faithfully reflects the overall causal relationship between variables across all conditions. This is analogous to the distinction between informative and misinformative components in pointwise mutual information⁷⁸. As a result, SURD avoids overestimating causality based on rare or atypical states and provides a more robust and interpretable measure of causal influence.

For instance, consider a scenario where Q_1 has a strong influence on Q_j^+ when q_1 is a rare state that occurs only 1% of the time. In the remaining 99% of cases, knowledge of q_1 either does not affect or even reduces certainty about Q_j^+ . If we were to compute causality based solely on positive contributions, we might conclude that Q_1 is a significant causal factor for Q_j^+ due to the strong influence in that 1% of cases. However, this would be misleading, as in 99% of the instances, Q_1 does not provide useful information about Q_j^+ . By including both causal and non-causal contributions in the SURD measure, we ensure that the overall causality reflects the true influence of Q_1 on Q_j^+ across all states. This is analogous to the distinction between informative and misinformative components in pointwise mutual information⁷⁸. As a result, SURD avoids overestimating causality based on rare or atypical states and provides a more robust and interpretable measure of causal influence.

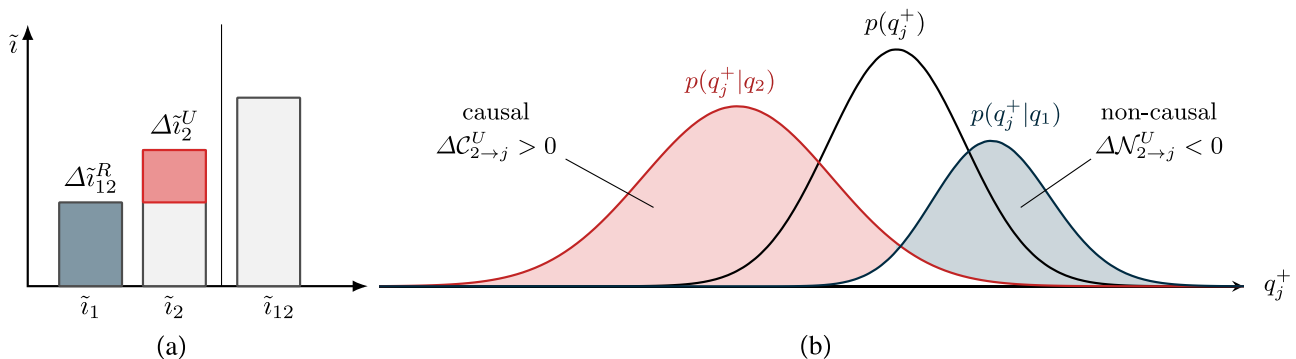


Fig. 6 | Interpretation of causal and non-causal states. a Ranking of specific mutual information between a target state q_j^+ and all combinations of two source variables $\mathbf{Q} = [Q_1, Q_2]$, namely $\tilde{I}_1, \tilde{I}_2$, and $\tilde{I}_{12}$. The redundant increment $\Delta\tilde{I}_{12}^R$ (in blue) and the unique increment $\Delta\tilde{I}_2^U$ (in red) are computed based on this ranking. **b** State-dependent unique causality $\Delta C_{2 \rightarrow j}^U$ (in red) and non-causality $\Delta \mathcal{N}_{2 \rightarrow j}^U$ (in blue) for a system with target variable Q_j^+ and source variables Q_1 and Q_2 , as defined by Eq. (6).

In this example, it is assumed that the specific mutual information satisfies $\tilde{I}_2 > \tilde{I}_1$ for all $q_j^+ \in Q_j^+$. Positive values of $\Delta C_{2 \rightarrow j}^U$ correspond to states where knowledge of q_2 increases the likelihood of the target state q_j^+ relative to q_1 , i.e., $p(q_j^+ | q_2) > p(q_j^+ | q_1)$, indicating unique causality. Conversely, negative or zero values (denoted by $\Delta \mathcal{N}_{2 \rightarrow j}^U$) identify non-causal states where q_2 provides less information about q_j^+ than q_1 , i.e., $p(q_j^+ | q_2) < p(q_j^+ | q_1)$.

Finally, it is worth noting that other approaches for decomposing mutual information into redundant, unique, and synergistic components have been proposed in the literature, such as PID⁵⁴ and its variants^{29,79–83}. While these approaches provide valuable insights into the structure of mutual information, it was discussed in ref. 53 that they either fail to satisfy all the properties required by our method or result in an unmanageable number of terms. Hence, our formulation does not follow PID, and the meanings of redundancy, uniqueness, and synergy adopted in this work differ from those in PID and related approaches. Instead, our framework focuses on quantifying the increments of information gained about the states of the target variable from different combinations of source variable states. Although this definition departs from earlier notions of redundancy, synergy, and uniqueness⁵⁴, it enables us, for example, to group synergistic components by specific order or to account for them through the concept of causality leak, thanks to the additivity of causal components. In our view, this simplifies the interpretation of scenarios involving many variables, where the number of decomposed terms can become prohibitively large, or situations where the number of observations is very low. For instance, PID results in a number of terms that grows according to the Dedekind numbers; in the case of nine variables, this decomposition yields over 10^{23} terms, whereas our method produces 502 causal terms (along with 502 non-causal counterparts)—which, although still large, remains computationally feasible. Despite our different notion of redundancy, uniqueness, and synergy, by focusing on incremental information, our method ensures that causality is not double-counted and that the minimal set of variables containing information about the target states can be systematically identified. The reader is referred to the Supplementary Note 1.5 for a detailed comparison between our method and PID.

State-dependent conditional transfer entropy

We use a state-dependent version of CTE to quantify directed information transfer between a target variable Q_j^+ and a vector of source variables $\mathbf{Q}$. The CTE from a source variable Q_i to the target Q_j^+ is defined as:

$$\overline{\text{CTE}}_{i \rightarrow j} = H(Q_j^+ | \mathbf{Q}_i) - H(Q_j^+ | \mathbf{Q}), \quad (8)$$

where $\mathbf{Q}_i$ denotes the vector $\mathbf{Q}$ excluding the component Q_i , and $H(\cdot | \cdot)$ is the conditional Shannon entropy²⁷. This expression measures the additional information that Q_i provides about Q_j^+ beyond what is already available from the remaining variables in $\mathbf{Q}$. By expanding the conditional Shannon entropies in the definition above, we obtain a state-level representation of CTE:

$$\text{CTE}_{i \rightarrow j} = \sum_{\mathbf{q}_{/i} \in \mathbf{Q}_{/i}} p(q_j^+, q_i, \mathbf{q}_{/i}) \log_2 \left(\frac{p(q_j^+ | q_i, \mathbf{q}_{/i})}{p(q_j^+ | \mathbf{q}_{/i})} \right). \quad (9)$$

For a system consisting of two source variables Q_1 and Q_2 , and a target variable Q_j^+ with joint probability distribution $p(q_j^+, q_1, q_2)$, the state-

dependent CTE from the source variable Q_2 to the target Q_j^+ can be expressed as:

$$\text{CTE}_{2 \rightarrow j} = \sum_{q_1 \in Q_1} p(q_j^+, q_1, q_2) \log_2 \left(\frac{p(q_j^+ | q_1, q_2)}{p(q_j^+ | q_1)} \right), \quad (10)$$

which compares the log of the conditional probabilities $p(q_j^+ | q_1, q_2)$ and $p(q_j^+ | q_1)$, weighted by the joint probability $p(q_j^+, q_1, q_2)$. In contrast to the specific unique causality defined in Eq. (6), we observe that here the comparison is made between $p(q_j^+ | q_1, q_2)$ and $p(q_j^+ | q_1)$, while in the unique causality the comparison is between $p(q_j^+ | q_2)$ and $p(q_j^+ | q_1)$.

Comparison with other methods

We compare the results for the two benchmark cases shown in Fig. 2 using five different methods that estimate time-varying measures of influence between variables: a state-dependent variant of CTE, local transfer entropy (LTE)⁸⁴, specific transfer entropy (STE)⁴⁴, local Granger causality (LGC)⁵², and TvLK information flow⁵⁰. LTE and STE are designed to quantify interactions among variables in dynamical systems without necessarily asserting an underlying causal structure, whereas LGC and TvLK are explicitly formulated as causal inference methods.

For LTE and STE, we employ the implementation by Darmon and Rapp⁴⁴, where LTE estimates are obtained using a k -nearest neighbors (k -NN) estimator with $k = 5$ neighbors⁸⁵. Following the authors' recommendations, the STE estimates are derived from LTE using $k_{\text{reg}} = \lfloor \sqrt{T} \rfloor$, where T denotes the total number of samples. This choice controls the degree of smoothing and balances the bias-variance trade-off in the estimation of STE. For LGC, we implement the formulation proposed in ref. 52 and validate our implementation by reproducing the toy examples presented in that study. For TvLK, we adopt the implementation described in ref. 50, applying a uniformly weighted moving average filter with a window length of 10 points, and a sliding estimation window of 100 points. We verified that varying the estimation window length between 30 and 150 points did not affect the conclusions. Further details on each method and their implementation are provided in the Supplementary Note 4.

We applied the five methods described above to the source- and target-dependent causality cases considered in the validation section. The results are visualized in Fig. 7, which shows the time evolution of q_1 (for source-dependent causality) and q_2^+ (for target-dependent causality). The signals are colored according to the value of CTE, LTE, STE, LGC, or TvLK. The expectation is that, for source-dependent causality, only instances where $q_1 > 0$ should exhibit strong causal influence from q_1 to q_2^+ , while for target-dependent causality, strong causal influence from q_1 to q_2^+ should be detected when $q_2^+ > 0$. However, the results reveal that none of the methods consistently exhibit the expected trends.

Among the methods evaluated, STE performs best in the source-dependent case, where larger values are observed for $q_1 > 0$, consistent with the dependence $q_1 \rightarrow q_2^+$ when $q_1 > 0$. However, STE remains nonzero

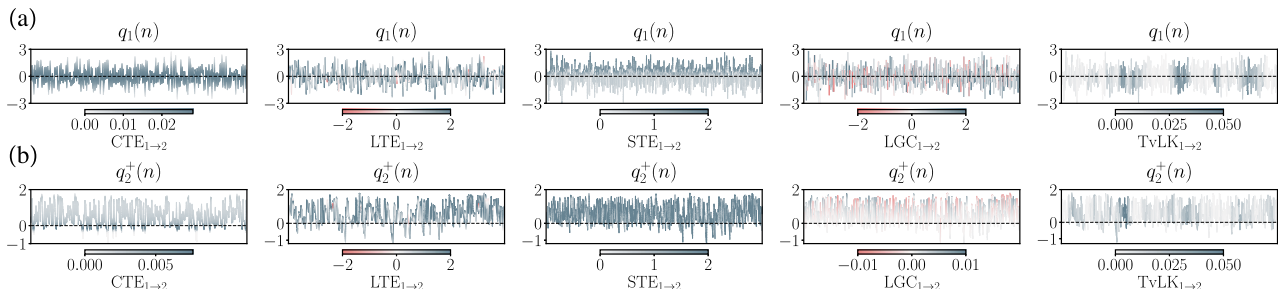


Fig. 7 | Comparison of time-varying causal inference methods. The approaches used include state-dependent conditional transfer entropy (CTE), local transfer entropy (LTE)⁸⁴, specific transfer entropy (STE)⁴⁴, local Granger causality (LGC)⁵²,

and time-varying Liang–Kleeman (TvLK) information flow⁵⁰. Results are shown for the two benchmark cases from Fig. 2: **a** system with source-dependent causality and **b** system with target-dependent causality.

even for $q_1 < 0$, where q_2 is also required to predict q_2^+ , indicating that STE does not fully resolve the dependence structure of the system. In the case of target-dependent causality, STE projects the results from LTE onto the future states of the target variable. As a consequence, the results for STE in the target-dependent case are not consistent with the functional dependencies inherent to the system.

LTE, LGC, and TvLK do not offer clear evidence for either source-dependent or target-dependent interaction. This outcome is not surprising for LGC and TvLK, as these methods are constrained by the assumptions of linearity and Gaussianity. In the present cases, the dependencies between variables are nonlinear, complicating the detection of the a priori causal dependencies. TvLK estimates time-varying causal relationships by applying a square-root Kalman filter to track the evolution of covariance matrices based on the original Liang–Kleeman information flow formulation. While this approach can be effective when the underlying probability distributions evolve smoothly over time, it may lack the sensitivity required to detect abrupt, short-time-scale changes in causality, such as those presented in the benchmark cases shown in Fig. 2.

Finally, in CTE, all redundant, unique, and synergistic causal contributions are aggregated into a single measure, obscuring the interpretation of the interactions between variables. For example, CTE fails to reveal in the source-dependent case that when $q_1 > 0$, the variable Q_1 uniquely determines the future of Q_2 , whereas when $q_1 < 0$, both Q_1 and Q_2 are jointly required to predict Q_2^+ . As a result, the detailed structure of variable interactions is lost when all forms of causality are conflated into a single quantity.

In closing this comparison, it is important to emphasize that the results produced by the methods discussed above are not incorrect, nor do we suggest otherwise. Each of these approaches offers valuable insights into interactions and information dynamics in complex systems, within the scope of their respective assumptions and intended applications, as demonstrated in prior studies. The key conclusion here is that these methods are not specifically designed to quantify the types of interaction and causality targeted in the present work. In contrast, our framework is explicitly tailored to address these challenges, particularly in systems characterized by source- or target-dependent causal structures.

We conclude this section by comparing the current method with our previous approach, SURD. The key similarities and differences are summarized in Table 1. Both SURD and the proposed method are grounded in the information-theoretic principle of forward-in-time information propagation. Similarly, both frameworks decompose causality according to the type of interaction—namely, redundant, unique, and synergistic contributions. However, SURD computes these causal contributions as averages over all system states, whereas the proposed method performs the decomposition at the level of individual states, both for the source and target variables. As a consequence, SURD may yield similar averaged causal attributions for systems with fundamentally different underlying state-dependent dynamics. This limitation is illustrated in Figs. 2 and 3, where source-dependent and target-dependent causal effects are discernible only through the proposed method. In addition, the current approach distinguishes between causal and non-causal components, with the latter corresponding to states whose knowledge increases uncertainty about the target. SURD does not offer this distinction. Another limitation of SURD is its inability to capture the temporal evolution of causality. In contrast, the proposed method allows for tracking such evolution, provided sufficient statistical support exists across the observed states. Overall, the current method enables a more detailed and state-resolved characterization of causal structure in complex systems, going beyond what is possible with existing causal inference frameworks.

Limitations

We discuss here some of the limitations of our method. First, the proposed causal framework is observational in nature—i.e., it infers causal relationships from statistical dependencies in time-resolved data without requiring interventions. While this allows for broad applicability to real-world systems

Table 1 | Comparison of SURD and the proposed method

Capability	SURD	Proposed method
Based on forward-in-time propagation of information	✓	✓
Interaction-specific decomposition (redundant, unique, synergistic)	✓	✓
State-dependent causality	×	✓
Causal vs. non-causal state distinction	×	✓
Temporal evolution of causality	×	✓

A checkmark (✓) indicates the feature is supported; a cross mark (×) indicates it is not. The table lists the main features supported by each framework.

where interventions are infeasible or unethical, it also introduces limitations: observational causality may not coincide with interventional or counterfactual definitions of causality, and can be confounded by hidden variables or latent dynamics.

Another key limitation is that the method is data-intensive, as it requires the estimation of probability distributions, which becomes increasingly challenging as the dimensionality of the dataset grows. This challenge primarily affects the quantification of synergistic causalities, which rely on higher-dimensional joint distributions compared to redundant and unique contributions. To mitigate this, our method permits restricting the order of synergistic interactions (e.g., limiting to pairwise synergistic causalities in systems with many source variables and limited data). The unestimated higher-order contributions can then be interpreted as a causality leak, thanks to the additive structure of our causal decomposition. Furthermore, recent advances in distribution estimation techniques, such as transport maps⁸⁶ and flow matching⁸⁷, can enhance the applicability of our approach to high-dimensional systems.

Another important consideration is that our method requires defining a partition of the state space of the system. In this work, we employed a uniform partition based on the values of the signals. However, in other contexts, phase-space partitioning may be tailored to the specific research question—for example, by distinguishing between regimes with and without extreme events. We believe that recent advances in quantized deep autoencoders, such as vector-quantized variational autoencoders⁸⁸ and finite scalar quantization methods⁸⁹, offer promising tools for constructing more efficient and data-adaptive state-space partitions.

Data for benchmark cases

The source-dependent and target-dependent benchmark cases comprise a system of two variables Q_1 and Q_2 at discrete times $t_n = n$. The system is initially set to $q_1(1) = q_2(1) = 0$. A stochastic forcing, represented by W_t , acts on Q_i and follows a Gaussian distribution. The computation of causalities is performed for a time lag of $\Delta T = 1$ using 75 uniform bins per variable. The integration of the system is carried out over 10^8 time steps, with the first 10,000 steps excluded from the analysis to avoid transient effects. In the analysis presented in Fig. 2, the causality leak for $q_2^+ = q_2(n+1)$ is 53% in the source-dependent case and 54% in the target-dependent case. This indicates that only 47% and 46% of the causality to Q_2^+ can be accounted for using Q_1 and Q_2 alone. The remaining unexplained part is due to the influence of W_2 , which also affects Q_2^+ but is not included in the analysis. An additional study with a system of three variables Q_1 , Q_2 , and Q_3 with similar functional relationships is reported in the Supplementary Notes 2.2, 2.3, 2.4, and 3.1, where we also discuss the structure of the state-dependent non-causal components.

Data for turbulent boundary layer

The data used for analyzing inner/outer interactions in a turbulent boundary layer were obtained for a dataset of a direct numerical simulation of the Navier–Stokes equations⁹⁰, where all temporal and spatial scales were resolved. The dataset covers the range of friction Reynolds number from $Re_\tau \approx 292$ –729, with 10,000 flowfields stored spanning 26 eddy-turnover

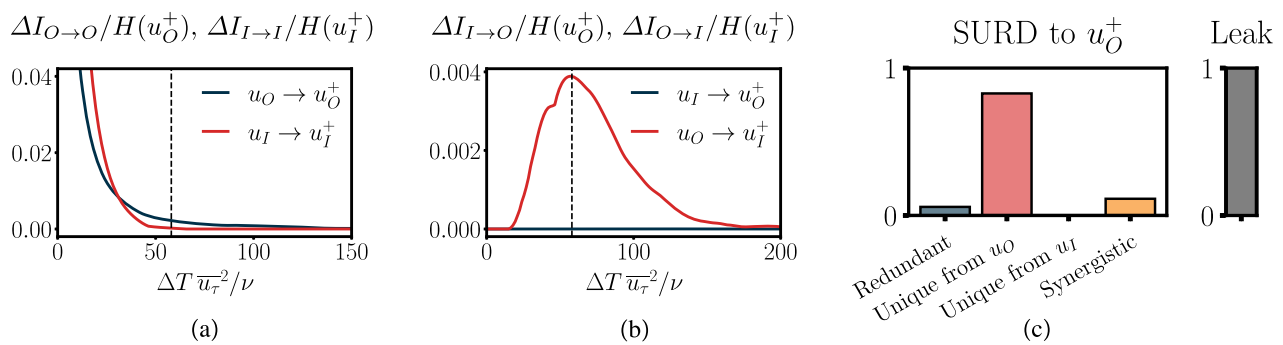


Fig. 8 | Selection of time lag for causal analysis in the turbulent boundary layer. **a** Self-induced ($\Delta I_{O \rightarrow O}^U$, $\Delta I_{I \rightarrow I}^U$) and **b** cross-induced ($\Delta I_{I \rightarrow O}^U$, $\Delta I_{O \rightarrow I}^U$) SURD causalities as a function of the time lag. Time is normalized using inner units based on the streamwise-averaged friction velocity $\bar{u}_\tau$ and viscosity ν and causalities with the Shannon information of the corresponding target variable, i.e. $H(u)$. The dashed

vertical line denotes the time lag selected for the causal analysis. **c** Redundant ($\Delta I_{O \rightarrow I}^R$), unique ($\Delta I_{I \rightarrow O}^U$), and synergistic ($\Delta I_{O \rightarrow I}^S$) causalities from SURD to the outer-layer motions, $u_O^+ = u_O(t + \Delta T)$ for the time lag of maximum cross-induced causality, i.e., $\Delta T = 60\nu/\bar{u}_\tau^2$.

times (after transients) based on $\bar{\delta}/\bar{u}_\tau$, where the symbol $\bar{(\cdot)}$ denotes the average along the streamwise direction x . Further details about the numerical setup can be found in Towne et al.⁶⁷. The data used in this study includes the temporal evolution of the streamwise velocity at two different wall-normal locations: $y_I = 4\nu/\bar{u}_\tau$ (for the inner layer) and $y_O = 0.3\delta$ (for the outer layer). The analysis was conducted at every streamwise location within the boundary layer, leveraging the self-similarity of the flow, which results in 2×10^7 samples for causal analysis. The time lag utilized to evaluate causality was $\Delta T = 60\nu/\bar{u}_\tau^2$, which corresponds to the time lag for maximum cross-induced unique causality, as shown in Fig. 8. At this time lag, the causality leak to the inner-layer velocity u_I^+ is 99%. This high value is expected, given that the vast majority of the degrees of freedom of the system (associated with the full turbulent flow field) are not accounted for ref. 53.

The temporal evolution of the self-induced causalities ($\Delta I_{O \rightarrow O}^U$ and $\Delta I_{I \rightarrow I}^U$) and cross-induced causalities ($\Delta I_{I \rightarrow O}^U$ and $\Delta I_{O \rightarrow I}^U$) is presented in Fig. 8. The causalities directed toward the outer-layer velocity u_O^+ at the same time lag are also shown in Fig. 8, revealing no significant contribution from the inner-layer velocity u_I . Finally, the probability distributions were discretized using 50 uniform bins per variable. Additional tests conducted with half and double the number of bins produced no significant differences in the results.

Data for climate science

The climate time series data are regional averages from the reanalysis for the period 1948–2012 with 780 months. WPAC denotes monthly surface pressure anomalies in the West Pacific, and CPAC and EPAC surface air temperature anomalies in the Central and East Pacific. Anomalies are taken with respect to the whole period, which is publicly available in <https://psl.noaa.gov/data/gridded/data.ncep.reanalysis.html>. The time lag utilized to evaluate causality is $\Delta T = 4$ months, which maximized the unique causality across different variables and is within the range of lags analyzed in Runge et al.¹⁶. At this time lag, the causality leak to WPAC($t + \Delta T$) and CPAC($t + \Delta T$) is 79% and 65%, respectively, which quantifies the effect from variables not included in the analysis. The joint distribution of the variables is estimated using a k -NN density estimator with $k = 7$ neighbors. Alternative values of k within the range [4,8] were tested, showing no significant impact on the results. To quantify uncertainty in the results, we performed 50 bootstrap resamplings on the available data. Additionally, to assess statistical significance, we applied 1000 random permutations to compute the p -values of the redundant and unique causalities reported in Fig. 5, both of which yielded p -values of zero. In contrast, when including synergistic causalities, the p -values increased significantly, leading us to conclude that the latter causalities lack sufficient statistical support with the available data. More details about the statistical significance of the results for this case are shown in the Supplementary Note 5.1, where we include the results for reduced sample sizes.

Data availability

The data generated in this study as well as the analysis and simulation code have been deposited in a Zenodo database under identifier <https://doi.org/10.5281/zenodo.17576116>.

Code availability

The codes developed for this work are available at: <https://github.com/ALD-Lab/SURD-states>.

Received: 15 September 2025; Accepted: 25 November 2025;

Published online: 16 December 2025

References

- Bunge, M. *Causality and Modern Science* (Dover Publications, 1979).
- Wu, X., Okumura, Y. M. & DiNezio, P. N. What controls the duration of El Niño and La Niña events? *J. Clim.* **32**, 5941 (2019).
- Cordes, D. et al. Mapping functionally related regions of brain with functional connectivity MR imaging. *Am. J. Neuroradiol.* **21**, 1636 (2000).
- Runge, J., Gerhardus, A., Varando, G., Eyring, V. & Camps-Valls, G. Causal inference for time series. *Nat. Rev. Earth Environ.* **4**, 487 (2023).
- Razi, A. & Friston, K. J. The connected brain: causality, models, and intrinsic dynamics. *IEEE Signal Process. Mag.* **33**, 14 (2016).
- Chiou-Wei, S. Z., Chen, C. F. & Zhu, Z. Economic growth and energy consumption revisited - evidence from linear and nonlinear Granger causality. *Energy Econ.* **30**, 3063 (2008).
- Rothman, K. J. & Greenland, S. Causation and causal inference in epidemiology. *Am. J. Public Health* **95**, S144 (2005).
- Hedström, P. & Ylikoski, P. Causal mechanisms in the social sciences. *Annu. Rev. Sociol.* **36**, 49 (2010).
- Lozano-Durán, A., Bae, H. J. & Encinar, M. P. Causality of energy-containing eddies in wall turbulence. *J. Fluid Mech.* **882**, A2 (2020).
- Martínez-Sánchez, Á. et al. Causality analysis of large-scale structures in the flow around a wall-mounted square cylinder. *J. Fluid Mech.* **967**, A1 (2023).
- Sugihara, G. et al. Detecting causality in complex ecosystems. *Science* **338**, 496 (2012).
- Ye, H., Deyle, E. R., Gilarranz, L. J. & Sugihara, G. Distinguishing time-delayed causal interactions using convergent cross mapping. *Sci. Rep.* **5**, 14750 (2015).
- Clark, A. T. et al. Spatial convergent cross mapping to detect causal relationships from short time series. *Ecology* **96**, 1174 (2015).
- Brouwer, E. D., Arany, A., Simm, J. & Moreau, Y. Latent convergent cross mapping. In *Proc. International Conference on Learning Representations* (2021).
- Javier, P. J. E. Causal-ccm: a Python implementation of Convergent Cross Mapping (2021).

16. Runge, J., Nowack, P., Kretschmer, M., Flaxman, S. & Sejdinovic, D. Detecting and quantifying causal associations in large nonlinear time series datasets. *Sci. Adv.* **5**, eaau4996 (2019).
17. Gerhardus, A. & Runge, J. High-recall causal discovery for autocorrelated time series with latent confounders. in *Advances in Neural Information Processing Systems* (eds Larochelle, H., Ranzato, M., Hadsell, R., Balcan, M. & Lin, H.) Vol. 33, 12615–12625 (Curran Associates, Inc., 2020).
18. Saggioro, E., de Wiljes, J., Kretschmer, M. & Runge, J. Reconstructing regime-dependent causal relationships from observational time series. *Chaos* **30**, 113115 (2020).
19. Brouillard, P. et al. Causal representation learning in temporal data via single-parent decoding. Preprint at <https://doi.org/10.48550/arXiv.2410.07013> (2024).
20. Granger, C. W. J. Investigating causal relations by econometric models and cross-spectral methods. *Econometrica* **37**, 424 (1969).
21. Murphy, K. A. & Bassett, D. S. Information decomposition in complex systems via machine learning. *Proc. Natl. Acad. Sci. USA* **121**, e2312988121 (2024).
22. Pearl, J. *Causality: Models, Reasoning, and Inference* (Cambridge University Press, 2000).
23. Eichler, M. Causal inference with multiple time series: principles and problems. *Philos. Trans. R. Soc. A* **371**, 20110613 (2013).
24. Tononi, G. & Sporns, O. Measuring information integration. *BMC Neurosci.* **4**, 31 (2003).
25. Hoel, E. P., Albantakis, L. & Tononi, G. Quantifying causal emergence shows that macro can beat micro. *Proc. Natl. Acad. Sci. USA* **110**, 19790 (2013).
26. Hoel, E. P., Albantakis, L., Marshall, W. & Tononi, G. Can the macro beat the micro? Integrated information across spatiotemporal scales. *Neurosci. Conscious.* **2016**, niw012 (2016).
27. Shannon, C. E. A mathematical theory of communication. *Bell Syst. Tech. J.* **27**, 379 (1948).
28. Landauer, R. The physical nature of information. *Phys. Lett. A* **217**, 188 (1996).
29. Lozano-Durán, A. & Arranz, G. Information-theoretic formulation of dynamical systems: causality, modeling, and control. *Phys. Rev. Res.* **4**, 023195 (2022).
30. Massey, J. Causality, feedback and directed information. In *Proc. 1990 International Symposium on Information Theory and its Applications* 27–30 (ACM, 1990).
31. Kramer, G. *Directed information for channels with feedback*, PhD Thesis, ETH Zürich, Zürich (1998).
32. Schreiber, T. Measuring information transfer. *Phys. Rev. Lett.* **85**, 461 (2000).
33. Verdes, P. Assessing causality from multivariate time series. *Phys. Rev. E* **72**, 026222 (2005).
34. Lizier, J. T., Prokopenko, M. & Zomaya, A. Y. Local information transfer as a spatiotemporal filter for complex systems. *Phys. Rev. E* **77**, 026110 (2008).
35. Barnett, L., Barrett, A. B. & Seth, A. K. Granger causality and transfer entropy are equivalent for Gaussian variables. *Phys. Rev. Lett.* **103**, 238701 (2009).
36. Lizier, J. T., Prokopenko, M. & Zomaya, A. Y. Information modification and particle collisions in distributed computation. *Chaos* **20**, 037109 (2010).
37. Bossomaier, T., Barnett, L., Harré, M. & Lizier, J. T. *An Introduction to Transfer Entropy: Information Flow in Complex Systems* 1st edn (Springer International Publishing, 2016).
38. Liang, X. S. & Kleeman, R. Information transfer between dynamical system components. *Phys. Rev. Lett.* **95**, 244101 (2006).
39. Liang, X. S. Information flow and causality as rigorous notions ab initio. *Phys. Rev. E* **94**, 052201 (2016).
40. Liang, X. S. Information flow within stochastic dynamical systems. *Phys. Rev. E* **78**, 031113 (2008).
41. Liang, X. S. The Liang-Kleeman information flow: theory and applications. *Entropy* **15**, 327 (2013).
42. Wang, W., Lozano-Durán, A., Helmig, R. & Chu, X. Spatial and spectral characteristics of information flux between turbulent boundary layers and porous media. *J. Fluid Mech.* **949**, A16 (2022).
43. Lizier, J. T. JIDT: An Information-Theoretic Toolkit for Studying the Dynamics of Complex Systems, *Front. Robot. AI* **1**, <https://doi.org/10.3389/frobt.2014.00011> (2014a).
44. Darmon, D. & Rapp, P. E. Specific transfer entropy and other state-dependent transfer entropies for continuous-state input-output systems. *Phys. Rev. E* **96**, 022121 (2017).
45. Martini, M., Kranz, T. A., Wagner, T. & Lehnertz, K. Inferring directional interactions from transient signals with symbolic transfer entropy. *Phys. Rev. E* **83**, 011919 (2011).
46. Wibral, M. et al. Measuring information-transfer delays. *PLoS ONE* **8**, e55809 (2013).
47. Wollstadt, P., Martínez-Zarzuela, M., Vicente, R., Díaz-Pernas, F. J. & Wibral, M. Efficient transfer entropy analysis of non-stationary neural time series. *PLoS ONE* **9**, e102833 (2014).
48. Gómez-Herrero, G. et al. Assessing coupling dynamics from an ensemble of time series. *Entropy* **17**, 1958 (2015).
49. Zhu, J. et al. A fast and efficient ensemble transfer entropy and applications in neural signals. *Entropy* **24**, 1118 (2022).
50. Hagan, D. F. T., Wang, G., Liang, X. S. & Dolman, H. A. J. A time-varying causality formalism based on the Liang-Kleeman information flow for analyzing directed interactions in nonstationary climate systems. *J. Clim.* **32**, 7521 (2019).
51. Zhou, F. et al. Estimating time-dependent structures in a multivariate causality for land-atmosphere interactions. *J. Clim.* **37**, 1853 (2024).
52. Stramaglia, S., Scagliarini, T., Antonacci, Y. & Faes, L. Local Granger causality. *Phys. Rev. E* **103**, L020102 (2021).
53. Martínez-Sánchez, Á., Arranz, G. & Lozano-Durán, A. Decomposing causality into its synergistic, unique, and redundant components. *Nat. Commun.* **15**, 9296 (2024).
54. Williams, P. L. & Beer, R. D. Nonnegative decomposition of multivariate information. Preprint at <https://doi.org/10.48550/arXiv.1004.2515> (2010).
55. Stramaglia, S., Faes, L., Cortes, J. M. & Marinazzo, D. Disentangling high-order effects in the transfer entropy. *Phys. Rev. Res.* **6**, L032007 (2024).
56. Kořenek, A., Sanda, P. & Hlinka, J. Higher order definition of causality by optimally conditioned transfer entropy. *Phys. Rev. E* **111**, L042302 (2025).
57. Marusic, I., Mathis, R. & Hutchins, N. Predictive model for wall-bounded turbulent flow. *Science* **329**, 193 (2010).
58. Flack, K. A., Schultz, M. P. & Shapiro, T. A. Experimental support for Townsend's Reynolds number similarity hypothesis on rough walls. *Phys. Fluids* **17**, 035102 (2005).
59. Flores, O. & Jiménez, J. Effect of wall-boundary disturbances on turbulent channel flows. *J. Fluid Mech.* **566**, 357 (2006).
60. Hutchins, N. & Marusic, I. Evidence of very long meandering features in the logarithmic region of turbulent boundary layers. *J. Fluid Mech.* **579**, 1 (2007).
61. Mathis, R., Hutchins, N. & Marusic, I. Large-scale amplitude modulation of the small-scale structures in turbulent boundary layers. *J. Fluid Mech.* **628**, 311 (2009).
62. Busse, B. & Sandham, A. Parametric forcing approach to rough-wall turbulent channel flow. *J. Fluid Mech.* **712**, 169 (2012).
63. Mizuno, Y. & Jiménez, J. Wall turbulence without walls. *J. Fluid Mech.* **723**, 429 (2013).
64. Chung, D., Monty, J. P. & Ooi, A. An idealised assessment of Townsend's outer-layer similarity hypothesis for wall turbulence. *J. Fluid Mech.* **742**, R3 (2014).
65. Lozano-Durán, A. & Bae, H. J. Characteristic scales of Townsend's wall-attached eddies. *J. Fluid Mech.* **868**, 698 (2019).

66. Townsend, A. A. *The Structure of Turbulent Shear Flow* (Cambridge University Press, 1976).
67. Towne, A. et al. A database for reduced-complexity modeling of fluid flows. *AIAA J.* **61**, 2867 (2023).
68. Jiménez, J. & Pinelli, A. The autonomous cycle of near-wall turbulence. *J. Fluid Mech.* **389**, 335–359 (1999).
69. Jiménez, J. Cascades in wall-bounded turbulence. *Annu. Rev. Fluid Mech.* **44**, 27 (2012).
70. Bjerknes, J. Atmospheric teleconnections from the equatorial Pacific. *Mon. Weather Rev.* **97**, 163 (1969).
71. Cai, W. et al. Climate impacts of the El Niño–southern oscillation on South America. *Nat. Rev. Earth Environ.* **1**, 215 (2020).
72. Wunsch, C. Geophysical interplays: El Niño, La Niña, and the southern oscillation. *Science* **248**, 904 (1990).
73. Pan, X., Li, T. & Yu, J. Change of El Niño onset location around 1970. *npj Clim. Atmos. Sci.* **7**, 163 (2024).
74. Timmermann, A. et al. El niño–southern oscillation complexity. *Nature* **559**, 535 (2018).
75. Kullback, S. & Leibler, R. A. On information and sufficiency. *Ann. Math. Stat.* **22**, 79 (1951).
76. Kreer, J. A question of terminology. *IEEE Trans. Inf. Theory* **3**, 208 (1957).
77. DeWeese, M. R. & Meister, M. How to measure the information gained from one symbol. *Netw. Comput. Neural Syst.* **10**, 325 (1999).
78. Cover, T. M. & Thomas, J. A. *Elements of Information Theory* (Wiley-Interscience, 2006).
79. Griffith, V. & Koch, C. Quantifying synergistic mutual information. in *Guided Self-Organization: Inception* 159–190 (Springer, 2014).
80. Griffith, V. & Ho, T. Quantifying redundant information in predicting a target random variable. *Entropy* **17**, 4644 (2015).
81. Ince, R. A. Measuring multivariate redundant information with pointwise common change in surprisal. *Entropy* **19**, 318 (2017).
82. Gutknecht, A. J., Wibral, M. & Makkeh, A. Bits and pieces: understanding information decomposition from part-whole relationships and formal logic. *Proc. R. Soc. A* **477**, 20210110 (2021).
83. Kolchinsky, A. A novel approach to the partial information decomposition. *Entropy* **24**, 403 (2022).
84. Lizier, J. T. Measuring the dynamics of information processing on a local scale in time and space. in *Local Measures of Information Storage and Transfer in Complex Systems* 1–21 (Springer, 2014).
85. Kraskov, A., Stögbauer, H. & Grassberger, P. Estimating mutual information. *Phys. Rev. E* **69**, 066138 (2004).
86. Baptista, R., Marzouk, Y. & Zahm, O. On the representation and learning of monotone triangular transport maps. *Found. Comput. Math.* **24**, 2063–2108 (2023).
87. Fjelde, T., Mathieu, E. & Dutordoir, V. An introduction to flow matching <https://mlg.eng.cam.ac.uk/blog/2024/01/20/flow-matching.html> (2024).
88. Van Den Oord, A. et al. Neural discrete representation learning. In *Proc. 31st International Conference on Neural Information Processing Systems* 6309–6318 (ACM, 2017).
89. Mentzer, F., Minnen, D., Agustsson, E. & Tschannen, M. Finite scalar quantization: VQ-VAE made simple. in *The Twelfth International Conference on Learning Representations* (IEEE, 2024).
90. Towne, A. S. & Lozano-Durán, A. Turbulent boundary layer direct numerical simulations [Data set], University of Michigan - Deep Blue Data (2022).
91. DiLiberto, T., L’Heureux, M., Becker, E., Johnson, N. & Lindsey, R., The walker circulation: Enso’s atmospheric buddy, published on NOAA Climate.gov (2014).
92. Watanabe, M., Iwakiri, T., Dong, Y. & Kang, S. M. Two competing drivers of the recent walker circulation trend. *Geophys. Res. Lett.* **50**, e2023GL105332 (2023).
93. Runge, J. Discovering contemporaneous and lagged causal relations in autocorrelated nonlinear time series datasets. In *Proc. Conference on Uncertainty in Artificial Intelligence* 1388–1397 (PMLR, 2020).

Acknowledgements

The authors would like to thank Jakob Runge for providing the data for the climate science application and Gonzalo Arranz for his contributions to this work. This work was supported by the National Science Foundation under Grant No. 2140775 and MISTI Global Seed Funds, and UPM. Á.M.-S. received the support of a fellowship from the “la Caixa” Foundation (ID 100010434). The fellowship code is LCF/BQ/EU22/11930094. The authors acknowledge the MIT SuperCloud and Lincoln Laboratory Supercomputing Center for providing HPC resources that have contributed to the research results reported within this paper.

Author contributions

Á.M.-S.: Ideation, methodology, software, validation, investigation, data curation, writing—original draft, writing—review and editing, visualization. G.A.: Methodology, software, investigation, writing—review and editing. A.L.-D.: Ideation, methodology, writing—review and editing, supervision, resources, and funding acquisition.

Competing interests

The authors declare no competing interests.

Additional information

Supplementary information The online version contains supplementary material available at <https://doi.org/10.1038/s42005-025-02447-w>.

Correspondence and requests for materials should be addressed to Álvaro Martínez-Sánchez.

Peer review information *Communications Physics* thanks Julien Boussard and the other anonymous reviewer(s) for their contribution to the peer review of this work. A peer review file is available.

Reprints and permissions information is available at <http://www.nature.com/reprints>

Publisher’s note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Open Access This article is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License, which permits any non-commercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if you modified the licensed material. You do not have permission under this licence to share adapted material derived from this article or parts of it. The images or other third party material in this article are included in the article’s Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article’s Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit <http://creativecommons.org/licenses/by-nc-nd/4.0/>.

© The Author(s) 2025